Unit 50 Relationship between the structures and properties of polymers

Pre-study
A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

50.1 What decides the properties of a polymer?

ASL 2007 Paper 1 Q12

The properties of a polymer, such as its strength and flexibility, are decided by the characteristics of its molecules. For example, the stronger the intermolecular forces between the chains of a polymer, the stronger the polymer is.

In the following sections, we are going to look at the effect of structures and intermolecular forces on the properties of polymers.

50.2 High density polythene (HDPE) and low density polythene (LDPE)

Polymer chemists use the term crystalline to describe the areas in a polymer where the chains are closely packed in a regular way.

Many polymers contain mixtures of both **crystalline** (ordered) regions and **amorphous** (random) regions. In amorphous regions, the chains are further apart and have more freedom of movement. N1

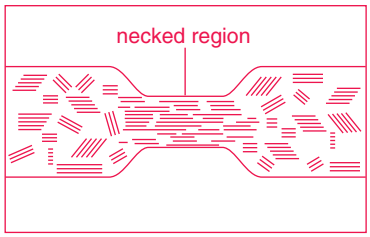
There are two main types of polythene, high density polythene (HDPE) and low density polythene (LDPE). High density polythene is stronger and more rigid than low density polythene.

Low density polythene contains relatively short chains with a considerable degree of branching, while chains of high density polythene are much longer and are not branched.

The lack of branching in high density polythene allows molecules to lie close together in a regular way and the crystallinity is over 90%.

ASL 2000 Q5(I)(d)

N1 Cold-drawing
All polymers contain some crystalline regions. When a polymer is stretched (cold-drawn) a neck forms. In the neck the polymer chains are lined up to form a more crystalline region. The process carries on until the whole polymer, except the ends which are being pulled, has aligned chains.



Cold-drawing leads to a larger increase in the polymer's strength. It is used to produce tough fibres.

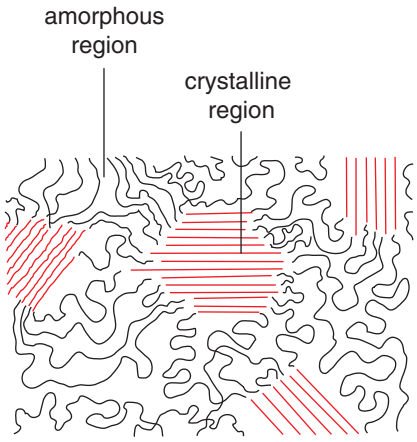


Fig. 50.1 Crystalline and amorphous regions of a polymer

crystalline 晶狀的　　amorphous 無定形的

Molecules of polythene are held together by van der Waals' forces. As the molecules of high density polythene lie close together, van der Waals' forces between the molecules are greater, and so the polymer is relatively strong and rigid. Its density is also higher due to the efficient packing of the molecules.

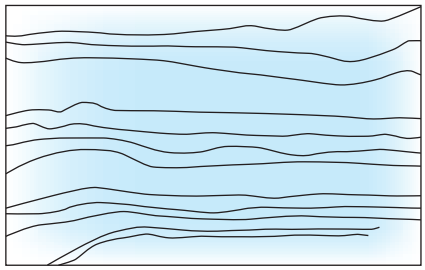


Fig. 50.2 Polymer chains in high density polythene (HDPE) are highly ordered

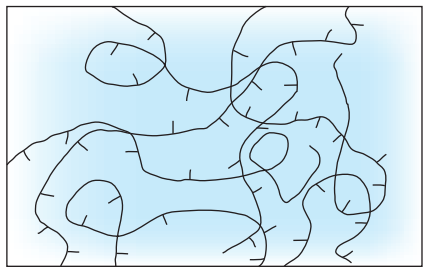


Fig. 50.3 Polymer chains in low density polythene (LDPE) contain many branches

On the other hand, the more branched chains of the low density polythene prevent the chains from lying close together in a tidy arrangement. Hence low density polythene has a significant proportion of amorphous regions.

Chains in amorphous regions are inefficiently packed, and so van der Waals' forces between the molecules are smaller. Hence low density polythene is softer, weaker and more easily deformed than high density polythene. The density of the polymer is also lower due to the space between the polymer chains.



low density polythene zig-zag bag

high density polythene bottle

Fig. 50.4 Different uses of polythene

Table 50.1 compares the properties of low density polythene and high density polythene.

Table 50.1 A comparison between LDPE and HDPE

	Low density polythene (LDPE)	High density polythene (HDPE)
Degree of branching from chains	high	low
Density (g cm^{-3})	0.91 – 0.94	0.95 – 0.97
Crystallinity	low (50% – 60% crystalline)	high (> 90% crystalline)
Tensile strength (MPa)*	5 – 15	20 – 40
Melting point (°C)	~115	~135

N2 Refer to page T35.

N2

* Tensile strength measures the ability of a material to withstand a stretching force. The unit used is megapascal MPa (where 1 MPa = 10^6 N m^{-2}).

50.3 Nylon and Kevlar

Nylon has found many uses in engineering because of its strength. Due to the presence of the N–H and C=O groups in the chains, hydrogen bonding* is possible between the chains. N3

* We have looked at the hydrogen bonding between nylon chains in Topic 8 Chemistry of Carbon Compounds.

N3 Hydrogen bonding between polymer chains is NOT a kind of cross linkage.

Fig. 50.5 shows part of the nylon-6,6 and Kevlar polymer chains. In nylon the single covalent bonds within the polymer chains are free to rotate and this makes polymer chains quite flexible. However, Kevlar is a polyamide containing benzene rings in the chains. The benzene rings make the polymer chains very rigid and linear.

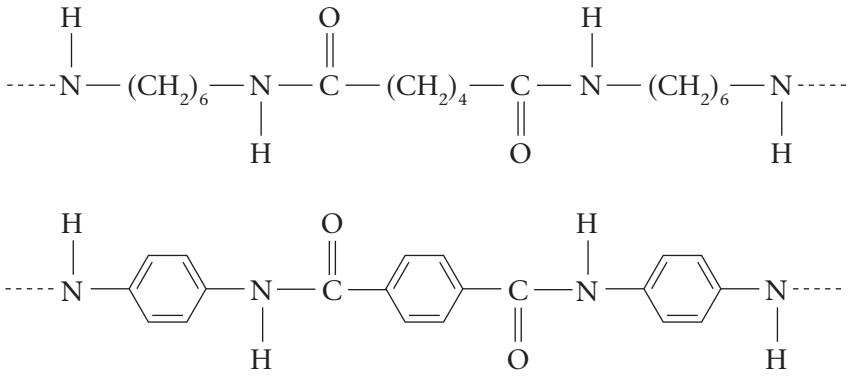
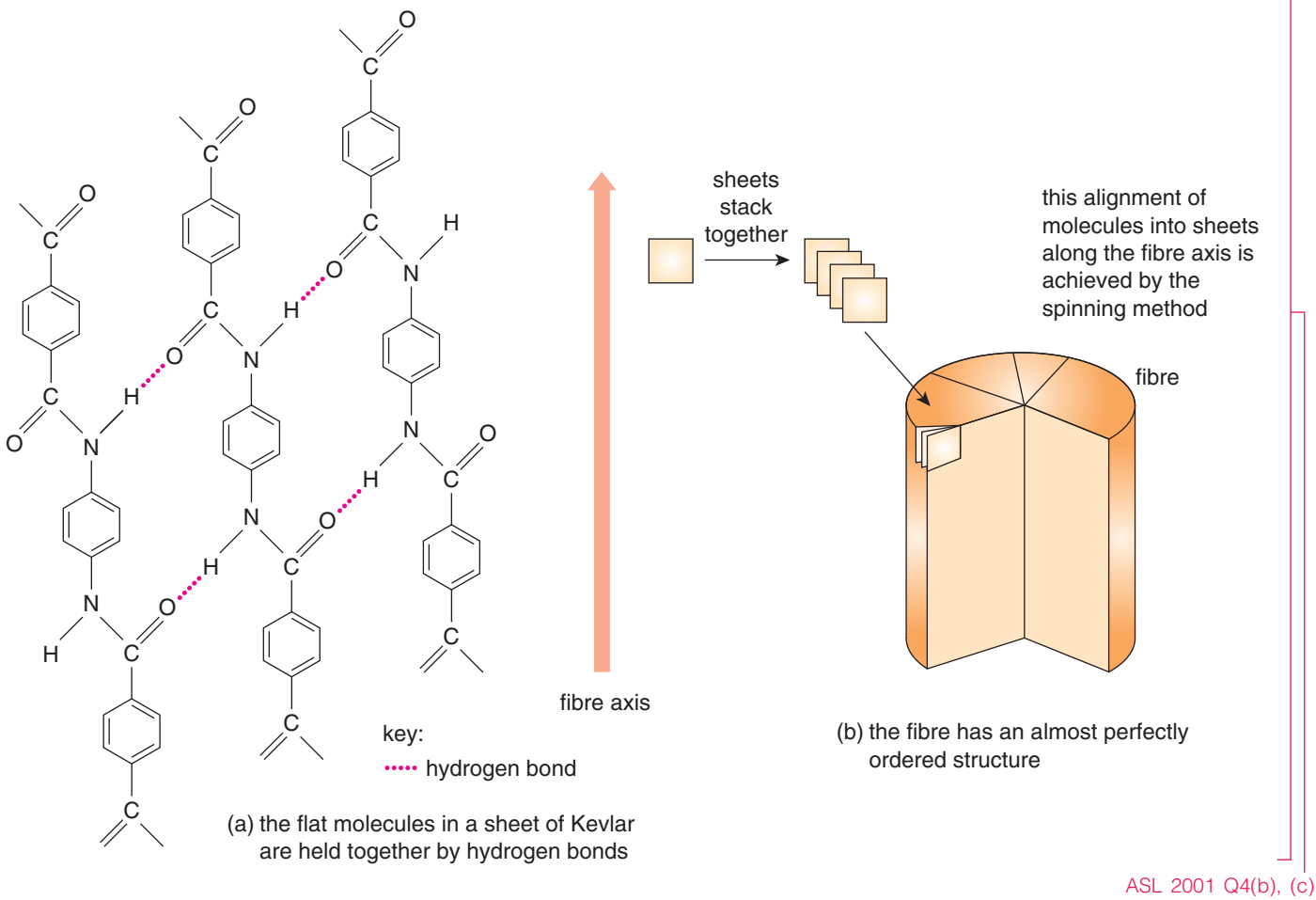


Fig. 50.5 Part of the nylon-6,6 and Kevlar polymer chains

Fig. 50.6a shows the hydrogen bonding in a sheet of Kevlar molecules. The hydrogen bonding causes the polymer chains to interlock with each other. The sheets then stack together in a regular fashion around the fibre axis to give a highly ordered structure. Hence the strength of Kevlar is greater than that of nylon. It is more rigid as well.



ASL 2001 Q4(b), (c)

Fig. 50.6 An illustration of the structure of Kevlar*

* All the $>C=O$ and $-N-H$ groups in the polymer chains are on opposite sides. This makes the chains highly symmetrical. This allows them to interlock with each other.

Table 50.2 compares the tensile strength of polypropene, nylon-6,6 and Kevlar.

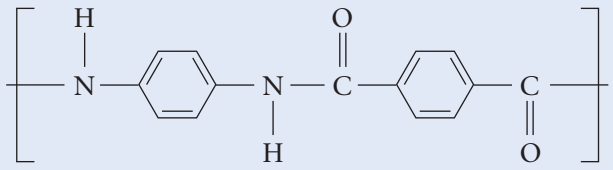
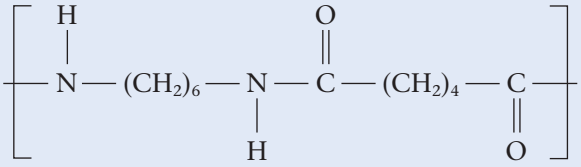
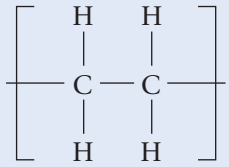
Table 50.2 Tensile strength of some common polymers

Polymer	Tensile strength (MPa)
Polypropene	31 – 41
Nylon-6,6	95
Kevlar	3 600 – 4 100

Example

50.1 Comparing the strength of three polymers

Consider the following three polymers, A, B and C. The repeating units of the polymers are shown in the table below.

Polymer	Repeating unit
A	
B	
C	

a) State the type of polymerization by which each polymer is formed.

b) The strength of the polymer is in the following order:

A > B > C

Explain this order in terms of the different types of intermolecular attractions between the polymer chains. N4

N4 Refer to page T35.

Solution

a)

Polymer	Type of polymerization
A	condensation
B	condensation
C	addition

DSE 2012 Paper 2 Q2(a)(i)(2)

b) The following table shows the intermolecular forces between chains of the three polymers.

Polymer	Intermolecular forces between polymer chains
A	hydrogen bonds
B	hydrogen bonds N5
C	van der Waal's forces

N5 Refer to page T36.

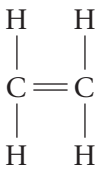
Both polymers A and B have hydrogen bonds between their polymer chains. While only weaker van der Waals' forces exist between the chains in polymer C. Hence polymers A and B are stronger than polymer C.

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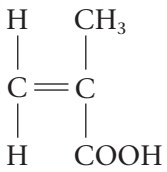
The chains in polymer A contain benzene rings. These rings make the polymer chains very rigid and linear. The hydrogen bonds cause the polymer chains to interlock with each other in sheets. The sheets then stack together to give a highly ordered structure. Hence polymer A has the greatest strength.

50.2 Examining the strength of a polymer

When ethene is allowed to react with 2-methylpropenoic acid under suitable conditions, a polymer X is formed. X does not have a definite repeating unit.

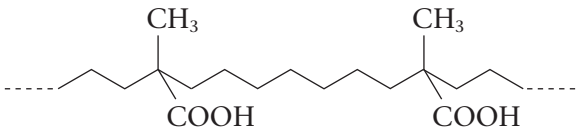


ethene



2-methylpropenoic acid

A part of the structure of X is shown below:



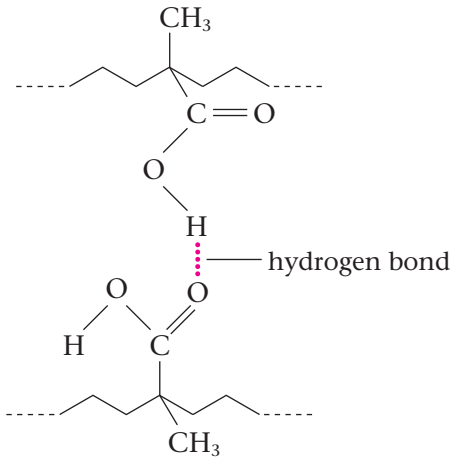
a) Hydrogen bonds exist between the chains of polymer X. Draw a diagram to show that.

b) Explain why the strength of X obtained depends on the relative amounts of ethene and 2-methylpropenoic acid used in the polymerization.

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Solution

a)



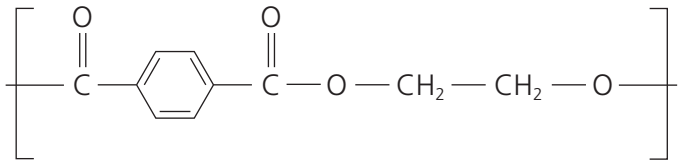
ASL 2002 Q12(b)(I)(ii)(4)
AL 2004 Paper 2 Q7(a)(iii)

b) Increasing the 2-methylpropenoic acid to ethene ratio increases the number of hydrogen bonds per unit length. The attraction between the polymer chains increases. Hence the strength of the polymer increases.

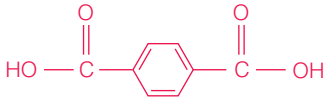
Checkpoint

1 Non-returnable drinks bottles are often made from PET.

a) PET is a polyester. The repeating unit of PET is shown below.



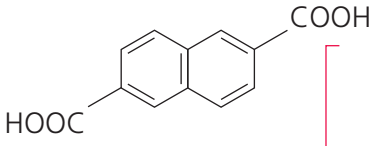
Draw the structural formulae of the monomers used in making PET.



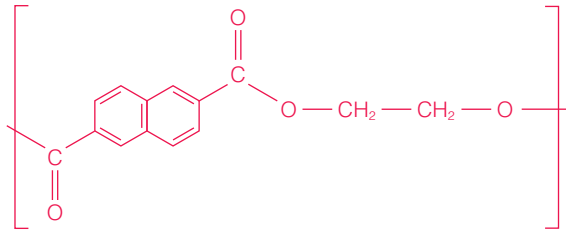


1 b) ii) The polymer chains of PEN contain flat benzene rings which prevent the chains from rotation, making the chains very rigid. Thus, the polymer chains can pack more regularly.
Hence the intermolecular forces between the polymer chains of PEN are stronger than those between the polymer chains of PET.

b) Chemists have recently developed a new polyester which can be used for producing returnable bottles. It can be made from ethane-1,2-diol and compound X. The structure of compound X is shown below. The polymer is known as PEN.



Compound X



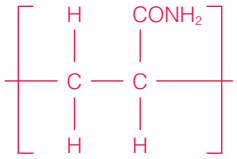
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AL 2007 Paper 2 Q6(d)(ii)(I)

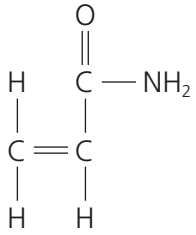
i) Draw the repeating unit of PEN.

ii) PEN has a higher melting temperature than PET with a similar relative molecular mass. Suggest why PEN has a higher melting temperature than PET.

2 The structure of acrylamide is shown below:

a)



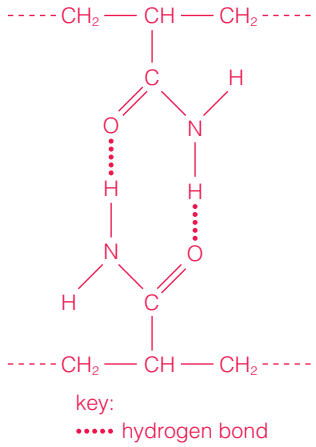


Acrylamide undergoes addition polymerization to form polyacrylamide.

a) Write the repeating unit of this polymer.

b) Draw a diagram to show how the polymer chains in polyacrylamide are attracted to each other.

b)



c) Hence explain why polyacrylamide has a high strength.
Hydrogen bonds exist between the polymer chains of the polymer.

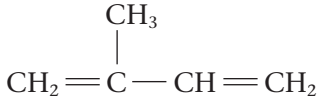


Fig. 50.7 Natural rubber is obtained from the milky, white liquid extracted from the rubber tree

latex 膠漿

50.4 Vulcanization of polymers

Natural rubber is a natural polymer. It is obtained from the milky, white liquid called *latex* extracted from the rubber tree. Natural rubber is formed by the addition polymerization of 2-methylbuta-1,3-diene, commonly called isoprene.



2-methylbuta-1,3-diene
(isoprene)

When the diene undergoes addition polymerization, one of the double bonds is retained in the resulting polymer (Fig. 50.8).

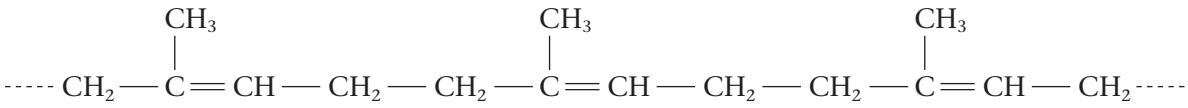
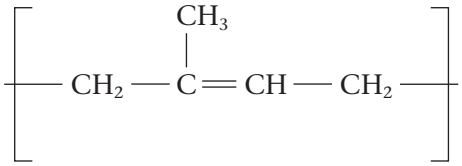


Fig. 50.8 Structure of natural rubber

The repeating unit is

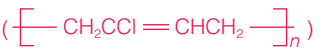


N6 Action of bromine on natural rubber gloves

A technician wore a pair of gloves made with natural rubber when performing an experiment. He spilt some bromine onto the gloves and the gloves became brittle.

Bromination reaction occurs to the C=C bonds of the natural rubber. The brominated polymer has a weaker intramolecular structure. This leads to a change in mechanical property.

N7 Vulcanization of a synthetic rubber

Neoprene () is a synthetic rubber. Its strength can be improved after heating it with sulphur. This is because cross-linkages are formed between the polymer chains, restricting the motion between the chains.

N6

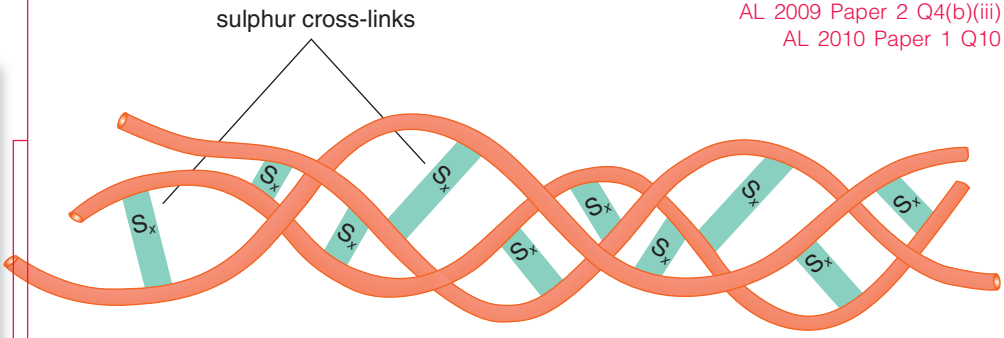
Elasticity is the ability of a material to regain its former shape after a distorting force is removed. Natural rubber is soft and has a poor strength and elasticity. **Vulcanization*** (heating natural rubber with sulphur) greatly improves these properties. The sulphur reacts with some of the double bonds in each chain, forming cross-links of one to about four sulphur atoms between the chains. This three-dimensional cross-linked structure makes vulcanized rubber a harder, stronger substance that is suitable for car tyres. N7

ASL 2010 Paper 1 Q12
AL 2009 Paper 2 Q4(b)(iii)
AL 2010 Paper 1 Q10

* Vulcanization originally referred to the cross-linking of rubber polymers. It has now expanded to include the cross-linking of all commercial polymers.



Fig. 50.10 Vulcanized rubber is used for tyres



where the subscript *x* indicates a variable number of sulphur atoms, usually 1 to 4, that form the cross-links

Fig. 50.9 Chains in natural rubber are cross-linked by vulcanization

Cross-linking also improves the elasticity of rubber. Since the chains have strong covalent bonds linking them, they cannot slide past each other when they are pulled. They return to their original positions after a distorting force is removed (Fig. 50.11).

DSE 2012 Paper 2 Q2(b)(i)

vulcanization 硫化作用

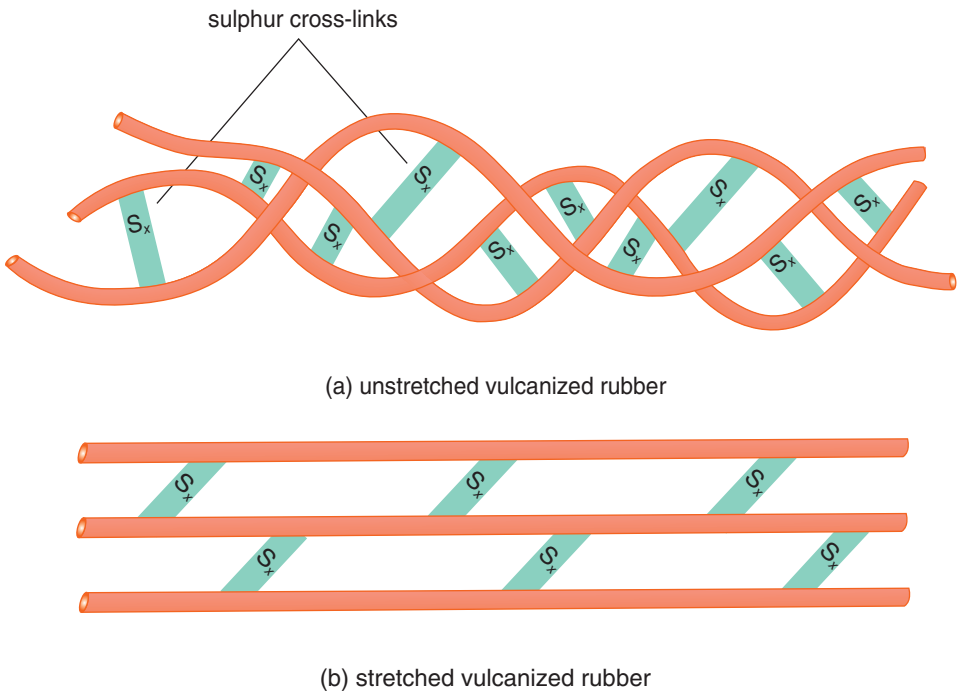


Fig. 50.11 Vulcanized rubber (a) unstretched (b) stretched



Fig. 50.12 Surgical gloves are produced from vulcanized rubber

Table 50.3 lists several types of rubber products and the number of monomer units between cross-links of the polymer chains. The smaller the number of monomer units between cross-links, the greater is the extent of cross-linking. The harder items have more cross-links holding the polymer chains together.

Table 50.3 Number of monomer units between cross-links of polymer chains in different rubber products

Product	Number of monomer units between cross-links
Surgical gloves	100 – 150
Kitchen gloves	50 – 80
Bicycle tyre	10 – 20

Today, the most common synthetic rubber used for making car tyres is a polymer of two different monomers: styrene and butadiene in a 1 : 3 ratio.

50.5 Degradable plastics

Degradation of a plastic is defined as a harmful change in its appearance, physical properties or chemical structure.



Fig. 50.13 An umbrella bag with additives added to make it degradable

In recent years research has focused on developing plastics that are degradable but with properties similar to conventional petroleum-based plastics. Two major directions to this research are:

- modifying petroleum-based plastics with additives; and
- making plastics from plant-based materials.

In the development of degradable plastics, adding additives is a popular strategy due to price competition.

Plastics prepared from plant-based materials are desirable because

- the monomers come from renewable sources such as corn, wheat, rice, etc.;
- as the monomers are natural materials, the monomers and polymers can be broken down by microorganisms, i.e. biodegradable.

Biodegradable plastic is a degradable plastic in which the degradation results from the action of naturally occurring microorganisms, such as bacteria, fungi, etc.

DSE 2014 Paper 2 Q2(c)(iv)(2)
AL 2004 Paper 1 Q10

* We have discussed polylactide in Unit 49.

Production process of polylactide (PLA) — a biodegradable polyester

ASL 2010 Paper 1 Q12
AL 2010 Paper 1 Q10

Polylactide*(PLA) is a polyester whose monomer is lactic acid. The lactic acid can be obtained from the petrochemical industry but its cost is very high. An environmentally friendly alternative is to obtain lactic acid from starch* from renewable resources, such as corn. N8

* Starch is a polysaccharide made up of glucose monomer molecules.

N8 Massive production of PLA may affect food supply.

The production of polylactide from starch involves the following main steps:

degradation 降解

AL 2004 Paper 1 Q10

Step 1 conversion of starch to glucose;

Step 2 fermentation* of the glucose to form lactic acid;

Step 3 conversion of two lactic acid molecules into a lactide;

Step 4 conversion of the lactide into the PLA polymer by polymerization.

* Fermentation is the slow breakdown of large organic molecules into small molecules by microorganisms in the absence of oxygen. We have discussed this in Topic 10 Rate of Reaction.

Fig. 50.14 shows the production process of polylactide from corn. Through a condensation reaction, two lactic acid molecules are converted into one cyclic molecule called a lactide. A solvent-free polymerization process causes the ring-shaped lactide to open and join end-to-end to form long polymer chains.

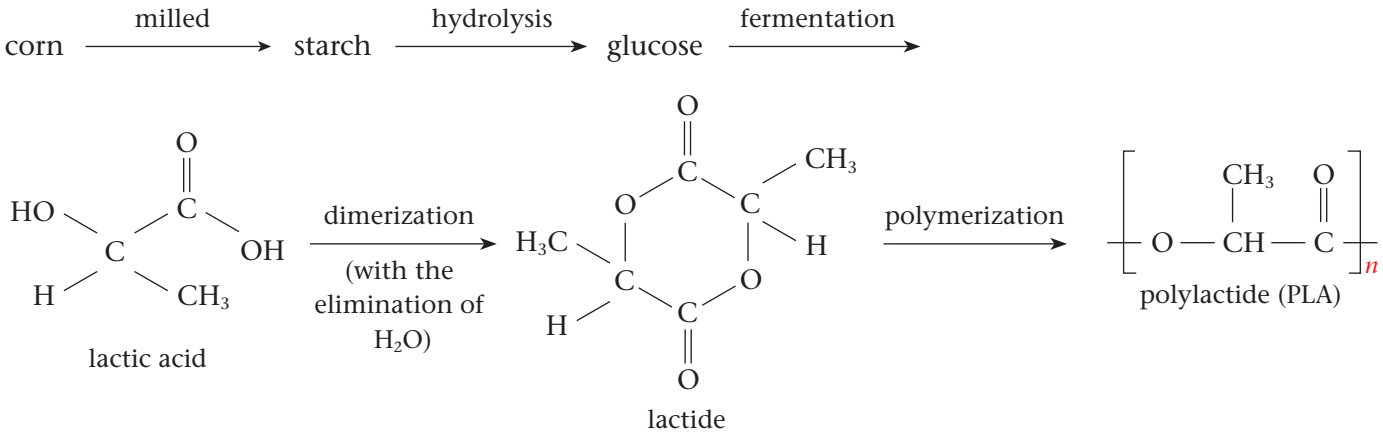


Fig. 50.14 Production process of polylactide from corn

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AL 2004 Paper 1 Q10

The production of PLA consumes 20 – 50% less fossil fuels* than comparable petroleum-based plastics.

In damp soil, PLA is decomposed by microorganisms, without leaving any residue or giving rise to toxic by-products. In commercial *composting* facilities, complete decomposition of PLA can be achieved within 30 – 50 days.

* Energy is required for the production of fertilizers, transport of raw materials, polymer production process, etc. Today, the energy is generated primarily from fossil fuels.



Fig. 50.15 Decomposition of a PLA cup

composting 堆肥

Uses of polylactide

Polylactide is a transparent, stiff thermoplastic. It is resistant to moisture and grease. It has flavour and odour barrier characteristics as well.

Major applications of polylactide include packaging materials and fibre for clothing. It can be used to make bottles, candy wraps, disposable plates and cups, etc..



Fig. 50.16 PLA bottles



Fig. 50.17 PLA cups



Fig. 50.18 Polylactide threads are used for surgical stitches

N9 Polyesters can be broken down via the hydrolysis of the ester linkage by enzymes or microorganisms.

Polylactide is also suitable for sporting products, such as golf clothes, due to its low moisture absorption and high *wicking* properties.

AL 1999 Paper 2 Q5(c)(iii)

Polylactide can be broken down by the hydrolysis of the ester linkage. Surgeons use polylactide threads to stitch together the sides of wounds. The polymer forms strong threads, but water in the body slowly hydrolyzes the ester linkage. The hydrolysis product is lactic acid which is non-toxic. N9

Adding a natural filler to polythene

Using a natural filler that can be attacked by microorganisms can also speed up the degradation of polythene. For example, some plastic bags are made of polythene which has starch granules added to it (Fig. 50.19). The starch is digested by microorganisms in the soil when the plastic bag is buried. The bag then breaks up into very small pieces of polythene. The very small pieces left over have a large surface area. This greatly speeds up their degradation.

AL 2004 Paper 1 Q10
AL 2010 Paper 1 Q10

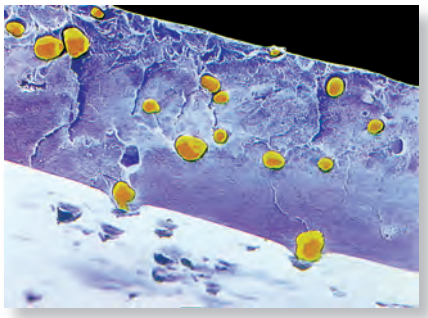


Fig. 50.19 Starch granules are added to the plastic

wicking 排汗

Photodegradable polythene

Carbonyl groups $-\overset{\overset{O}{\|}}{C}-$ absorb radiation in the wavelength range 270 nm – 360 nm*. This corresponds to light in the near ultraviolet region of the spectrum. We can incorporate these groups into the polymer chains to act as energy traps. The trapped energy causes the fission of bonds in the neighbourhood of the carbonyl groups, and the polymer chain breaks down into short fragments. This speeds up their degradation greatly.

* We will discuss the interaction of radiation with matter in Topic 16 Analytical Chemistry.

For example, carbonyl groups can be incorporated into polythene chains during the manufacturing process (Fig. 50.20).

AL 2004 Paper 1 Q10
AL 2010 Paper 1 Q10

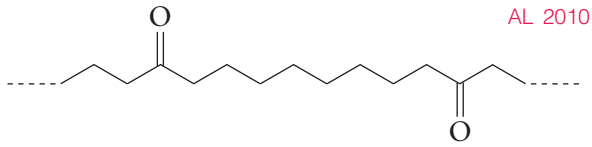


Fig. 50.20 A polythene chain with carbonyl groups incorporated

Status of degradable plastics today

At present, degradable plastics are not widely used. Some reasons are listed below:

1 Degradable plastics are much more expensive than commonly used plastics.

2 Control over the rate of degradation is necessary. For example, articles made of photodegradable polymer may be stored or used under different conditions. They may begin to degrade before the right time, causing inconvenience or even safety concerns.

3 When buried in landfills, degradable plastic articles will not be exposed to sunlight and sufficient air supply. They may remain unchanged for a long time, just like other plastics.

4 Little is known about the long term effects of residues of degradable plastic articles on the environment.

5 Degradable plastic articles look like other plastic articles. There is no quick and efficient method for their separation. If the two types of article are mixed up in the recycling process, articles made from the recycled plastic may become weaker.

Worksheet

Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.

STSE Connections

Biopol — the world's first bacterial polymer produced commercially

Polyesters play a predominant role as biodegradable polymers due to their hydrolyzable ester linkages. Besides polylactide (PLA), Biopol is another biodegradable polyester. It is a biopolymer made by certain bacteria. Microorganisms commonly found in the soil and natural water sources can break down this polymer under the anaerobic condition of a landfill.



Fig. 50.21 Biopol bottles at stages of increasing biodegradation

Polyhydroxybutyrate (PHB)

Polyhydroxyalkanoates (PHAs) are polyesters naturally produced by bacteria on sugar-based medium. Most research of PHAs has been on **polyhydroxybutyrate** (PHB), a short chain polymer produced by the bacterium *Alcaligenes eutrophus*.

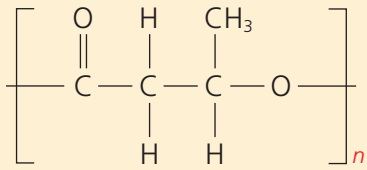


Fig. 50.22 Structure of polyhydroxybutyrate (PHB)

Unfortunately PHB has poor physical properties for commercial use. It is stiff, brittle and hard to process. This has led to an increased interest to produce copolymers with improved qualities. Polyhydroxyvalerate (PHV) is another main member of the PHA family. The copolymer PHBV (polyhydroxybutyrate valerate) is less stiff, tougher and easier to process, making it more suitable for commercial production. It is named Biopol. By growing Alcaligenes eutrophus in a high glucose, high valeric acid and low nitrogen environment, the bacteria make the PHBV copolymer, which they store in their cell walls as granules.

polyhydroxybutyrate 聚羥基丁酸酯　　Alcaligenes eutrophus 真養產鹼桿菌

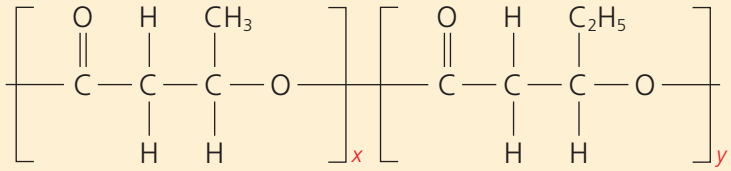


Fig. 50.23 Structure of PHBV

Biopol was first used as a shampoo bottle. The greatest demand was for packaging and disposable products. But the cost was too expensive to compete with conventional plastics and production ceased in 1998.

But we may not yet have seen the last of Biopol. Metabolix, a US biopolymer company, acquired the Biopol technology in 2001. The company produced a wide range of more useful PHAs using bacterial fermentation of renewable resources such as sugars and vegetable oils.

Questions Refer to page T36.

1 Discuss the advantages and disadvantages of using biodegradable polymers. You may visit the following website for some ideas.

"How green are green plastics?" — an article discussing the energy consumption and gas emission issues of polymers made from plants
http://www.mindfully.org/Plastic/Biodegrade/Green-PlasticsAug00.htm

2 In 2006, Metabolix announced that the company aimed to deliver a viable crop plant that generated economic yields of PHAs within five years. Search for information about the latest development of Metabolix on producing PHAs. Write a short report in less than 1 000 words to summarize your findings.

You may visit the website of Metabolix for some ideas.
http://www.metabolix.com/

N10 Refer to page T37.

50.6 Thermoplastics and thermosetting plastics N10

The structure of a thermoplastic

In a thermoplastic, such as high density polythene, the polymer molecules are in the form of separate long chains. Only weak intermolecular forces hold the polymer chains together.

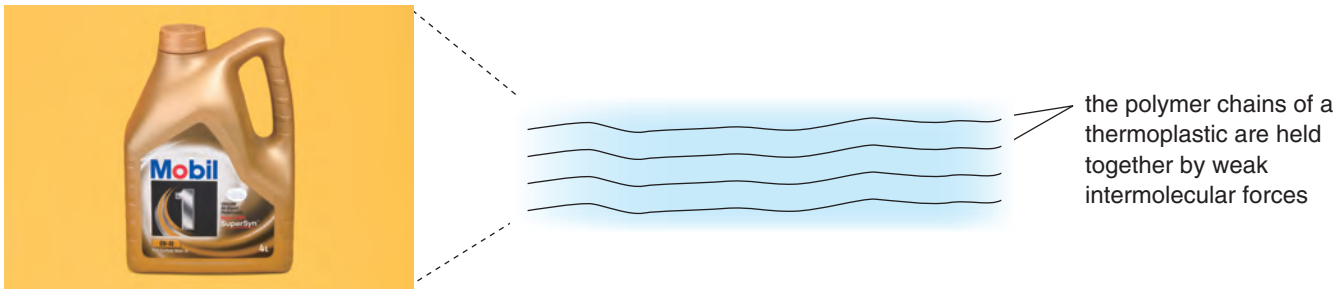


Fig. 50.24 The polymer chains of a thermoplastic are held together by weak intermolecular forces

CE 2003 Paper 2 Q36
CE 2006 Paper 1 Q11(a)
CE 2007 Paper 1 Q8(b)(i)
CE 2010 Paper 1 Q12(b)

Thermoplastics are relatively easy to soften and melt because a small amount of heat is needed to break the intermolecular forces. N11 When the plastic is cooled, the weak intermolecular forces are sufficient to cause the plastic to become solid again. Hence thermoplastics can be moulded or remoulded easily.

N11 Examination questions may ask students to explain why a certain polymer is a thermoplastic in terms of bonding and structure.

The structure of a thermosetting plastic

When a thermosetting plastic is made from its monomers, polymer chains are first formed. Hence it can be moulded into the desired shape.

As heating continues, strong links begin to form between the polymer chains to give a three-dimensional network of bonds. These links are called cross-links (Fig. 50.25). As a result, thermosetting plastics are set permanently in the shape they first form. Hence thermosetting plastics have to be moulded into shape when they are first made.

CE 2003 Paper 2 Q36
CE 2006 Paper 1 Q11(a)

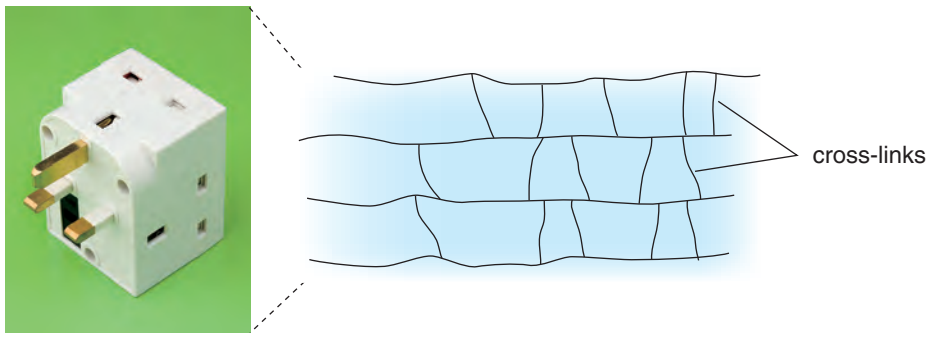


Fig. 50.25 The cross-links between the polymer chains of a thermosetting plastic make the plastic resistant to heat

AS 2012 Paper 1 Q5(b)
AL 2012 Paper 1 Q4(b)(ii)

Thermosetting plastics do not melt or soften on heating due to the cross-links between the polymer chains. Thermosetting plastics have many distinct properties, such as N12

- high thermal stability and insulating properties;
- hard and rigid;
- resistant to creep and deformation under load.

Fig. 50.26 summarizes information on the structures and properties of thermoplastics and thermosetting plastics. N13

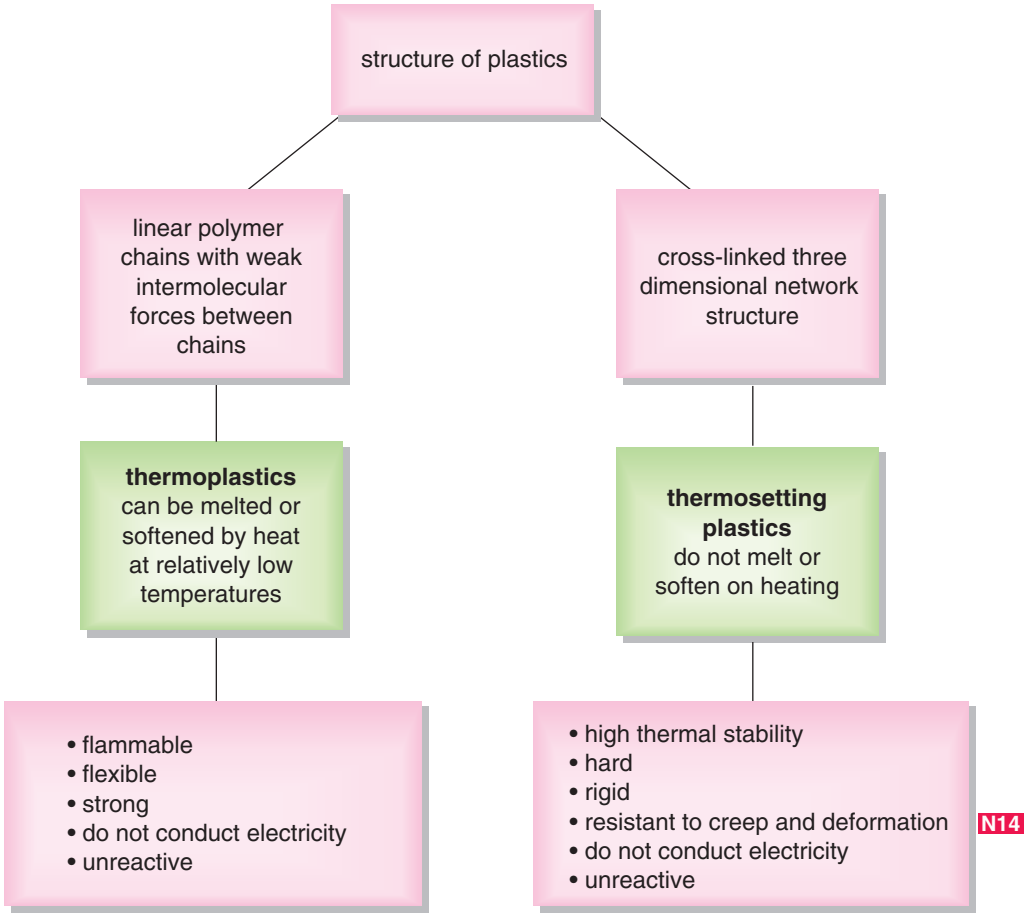


Fig. 50.26 A summary of the structures and properties of thermoplastics and thermosetting plastics

N12 Thermosetting plastic, upon strong heating, undergo decomposition rather than melting.

N13 Relating bonding and structure of polymers to their physical properties

Examination questions may ask students to discuss how bonding and structure of polymers affect their physical properties.

Bonding and structure

- Polymers are long chain compounds with very high molecular masses.
- Polymers with cross-links between polymer chains are thermosetting plastics, and those without cross-links are thermoplastics.
- In thermoplastics, the intermolecular attractions may be van der Waals' forces and / or hydrogen bonds. In thermosetting plastics, the cross-links are covalent bonds.
- Polymers can be classified as addition polymers and condensation polymers. Addition polymers are formed from monomers with C=C (or C≡C) bonds. Condensation polymers are formed from monomers each having two functional groups.

Physical properties

- The melting points of thermoplastics depend on the strength of intermolecular attractions, chain length and degree of branching from chains.
- Polymers with a higher degree of branching from chains have lower melting points. Those with a lower degree of branching have higher melting points (e.g. compare LDPE and HDPE).
- The rigidity of polymers can be reduced by the addition of plasticizers (e.g. PVC).
- Photodegradability of plastics can be improved by the introduction of light sensitive functional groups (e.g. C=O).

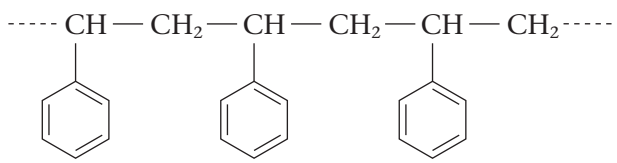
N14 Creep

Creep can be tested by suspending the polymer and hanging a weight on it. The polymer length is measured initially and at frequent time intervals during the test. The creep against time ratio is important.

Example

50.3 Studying the structure of polystyrene

Polystyrene is a plastic with a wide range of uses. It has the following structure:



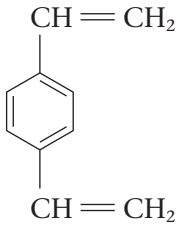
DSE 2015 Paper 2 Q2(b)(i)

ASL 2011 Paper 2 Q7(c)(i)

a) Draw the structure of styrene, the monomer of polystyrene. N15

b) Polymerization of styrene in the presence of 1,4-diethenylbenzene gives a thermosetting plastic P which is harder than polystyrene.

N15 Refer to page T38.



1,4-diethenylbenzene

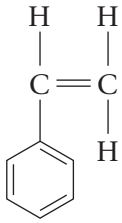
i) Draw part of the structure of plastic P.

ii) Explain why the hardness of plastic P depends on the relative amounts of styrene and 1,4-diethenylbenzene used.

ASL 2004 Q12(b)(II)(iii)
AL 2006 Paper 2 Q7(b)(ii)

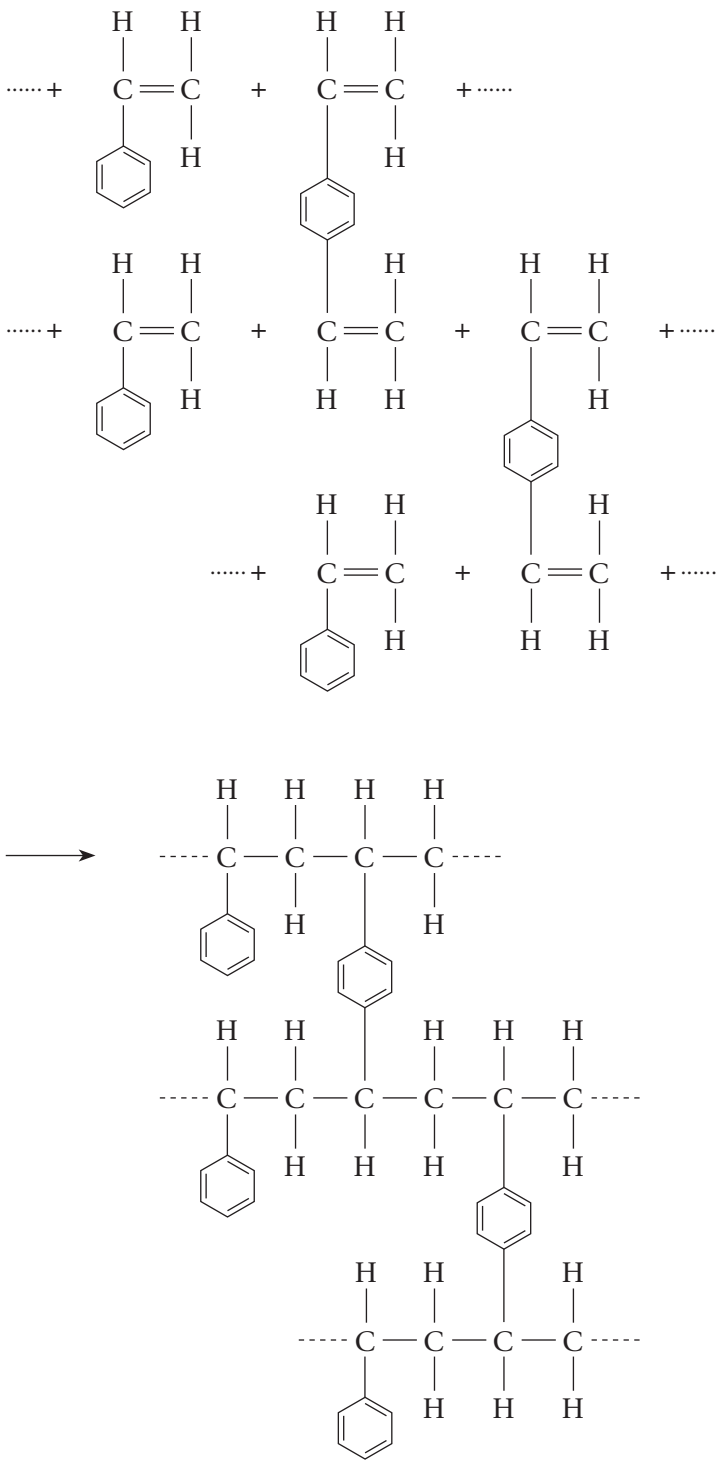
Solution

a)



ASL 2004 Q12(b)(II)(iii)
AL 2006 Paper 2 Q7(b)(ii)

b) i) The following diagram shows the polymerization that may occur during the polymerization of polystyrene in the presence of 1,4-diethenylbenzene:

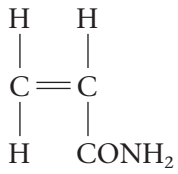


part of the structure of plastic P

ii) Styrene and 1,4-diethenylbenzene form a cross-linked thermosetting plastic P. The hardness of P increases with the 1,4-diethenylbenzene : styrene ratio (i.e. as the degree of cross-linking increases).

50.4 Accounting for the water absorbing property of a plastic

A brand of baby diaper uses plastic X as the water absorbing material. Plastic X is formed from the following monomer:

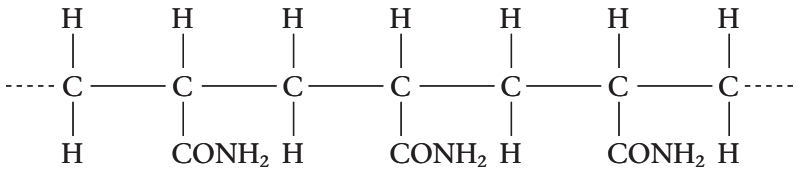


a) Draw part of the structure of plastic X.

b) Account for the water absorbing property of plastic X.

Solution

a)



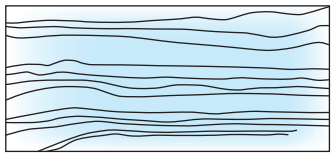
b) Plastic X contains a large number of amide groups ($-CONH_2$). These amide groups can form hydrogen bonds with water molecules.

ASL 2005 Q7(I)(a), (c)(i)

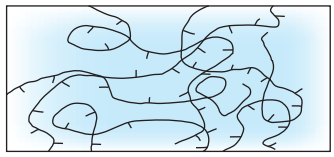


Checkpoint

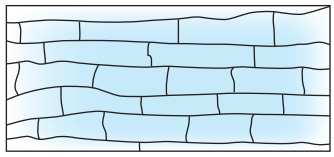
Answer the following questions by referring to the structures of three polymers shown below:



polymer A



polymer B



polymer C

Which one of these polymers

a) represents a soft, flexible polymer with a low softening temperature?
polymer B

b) might be used to make a laundry bucket?
polymer A

c) might be used to make the handles of a barbecue?
polymer C

50.7 Plastics fabrication processes

When designing a plastic product, the thermal properties of the plastic involved play an important role when designers decide which *fabrication process* to use.

As thermoplastics can be softened and solidified repeatedly, they are melted, shaped and cooled while maintaining the desired shape. Thermoplastic products can be remoulded*. Thermoplastics can be processed rapidly and are used to make inexpensive items.

* Scrap pieces can also be reheated and moulded again.

When a thermosetting plastic is first made from its monomers, it is in liquid form and can be moulded into the desired shape. As heating continues, cross-linking within the polymer occurs. A thermosetting plastic product cannot be remoulded. Thermosetting plastics are difficult to form, making their products costly.

We are going to discuss five plastics fabrication processes including

- **injection moulding**,
- **blow moulding**, CE 2010 Paper 1 Q12(d)
- **extrusion moulding**,
- **vacuum forming**, and
- **compression moulding**.

The first four processes are used primarily with thermoplastics while compression moulding is used mainly for thermosetting plastics.

AL 2010 Paper 1 Q10

Injection moulding

Injection moulding is a high-volume production technique. It is used predominantly for thermoplastics.

The injection moulding machine consists of an injection unit and a mould. The injection unit is an *extruder* with a screw that can be moved backwards and forwards. Plastic pellets are first fed into the top of the moulding machine. The material is melted by the heater. The melted plastic is then forced into the mould as the screw moves forwards. The plastic sets in the shape of the mould as it cools.

ASL 2000 Q5(I)(e)

fabrication process 加工過程　injection moulding 注塑法　blow moulding 吹塑法
extrusion moulding 擠塑法　vacuum forming 真空成型法　compression moulding 壓塑法
extruder 擠壓器

Reference Website

Refer to the following websites for animations showing the process of injection moulding:

- http://www.bpf.co.uk/plastipedia/processes/injection_moulding.aspx (accessed May 2015)
- http://www.youtube.com/watch?v=1-WjM0it9jI (accessed May 2015)

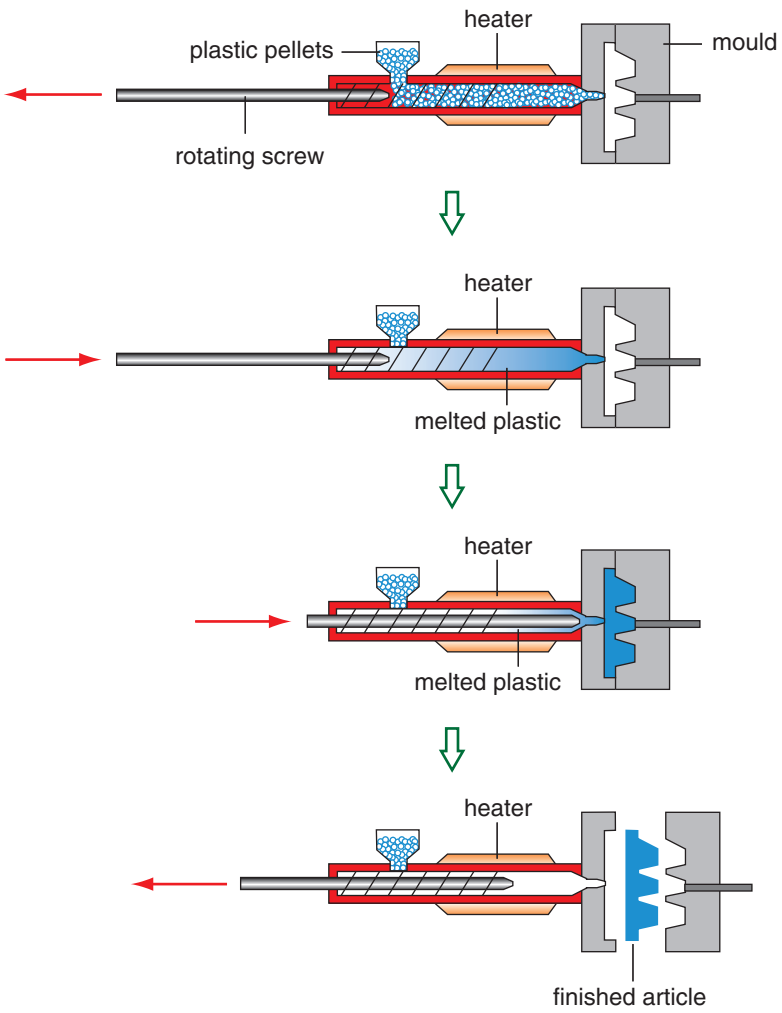


Fig. 50.27 Process of injection moulding

Injection moulding enables the mass production of complex shapes and very small items. Typical products fabricated by injection moulding include model kits, bottle crates, car bumpers, gear wheels, telephone cases, plastic chairs and washing-up bowls.



Fig. 50.28 Products fabricated by injection moulding

Reference Website
Refer to the following website for an animation showing the process of blow moulding:
http://www.notesandsketches.co.uk/Extrusion_Blow_Moulding.html
(accessed May 2015)

Blow moulding

Blow moulding is a fast method for manufacturing hollow objects like bottles. It is used for thermoplastics. The technique can also be used to produce large items, such as fuel tanks, toy tricycles and wheels.

DSE 2014 Paper 2 Q2(c)(i)

During the process, plastic pellets are fed from a hopper into an extruder. The pellets are melted by a heater and a short tube of melted plastic is extruded and trapped between the two halves of an open two-part mould (Fig. 50.29a). The mould is then closed and compressed air is blown into the plastic tube (Fig. 50.29b). The plastic then expands to take the shape of the inside of the mould, remaining hollow at its centre. The plastic sets in the shape of the mould as it cools (Fig. 50.29c).

ASL 1999 Q7(I)(c)

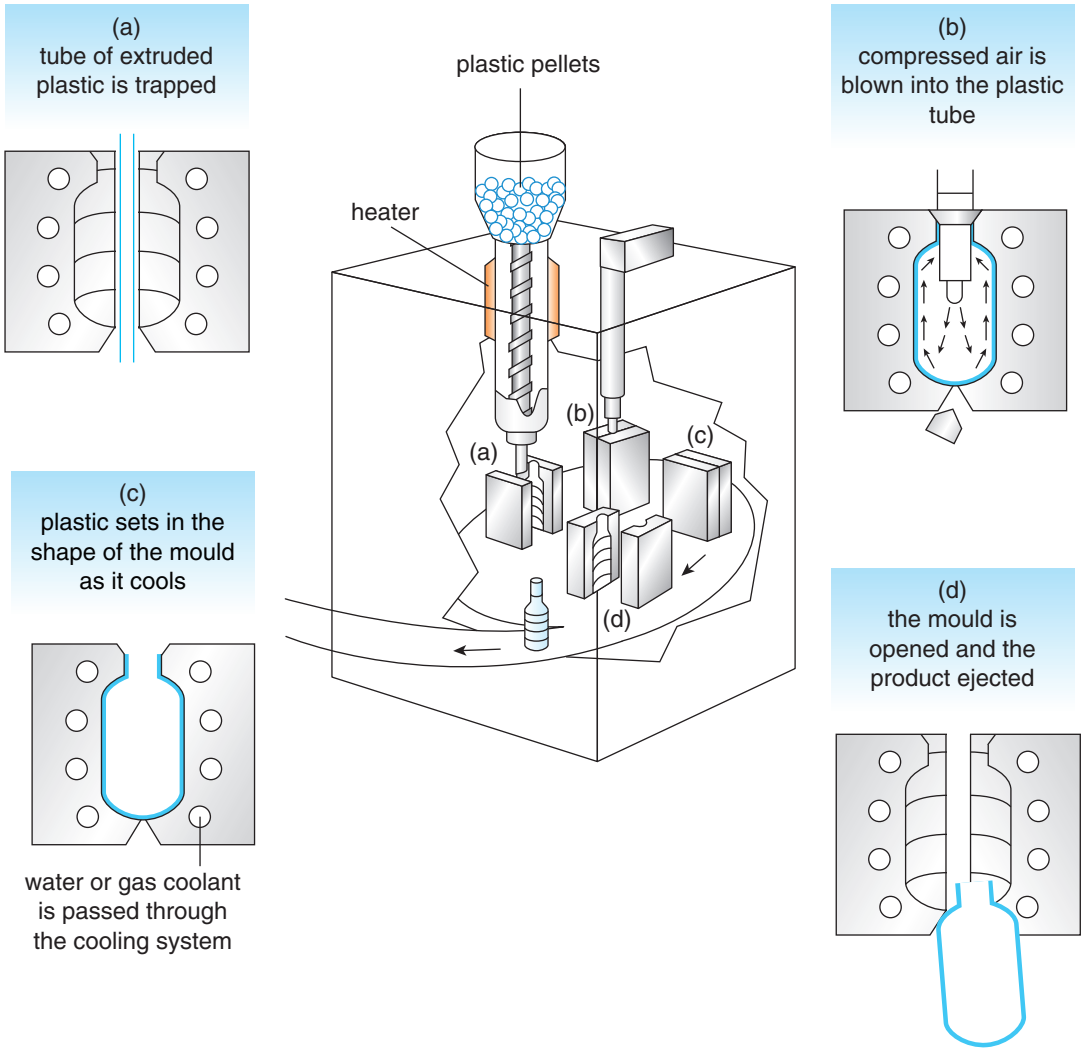


Fig. 50.29 Blow moulding



Fig. 50.30 Products fabricated by blow moulding

Typical products fabricated by blow moulding include bottles made from polyvinyl chloride (PVC), polypropene (PP), or high density polythene (HDPE), petrol tanks and drums.

Extrusion moulding

ASL 2006 Q13(b)(I)(ii)(I)

Extrusion moulding is employed to form thermoplastic materials into continuous sheeting, film, tubes, rods and filaments, and to coat wires, cables and cords.

In extrusion, plastic pellets are loaded into a hopper, then fed into a long heating chamber. At the end of the heating chamber, the molten plastic is forced out through a specially shaped opening called a die.

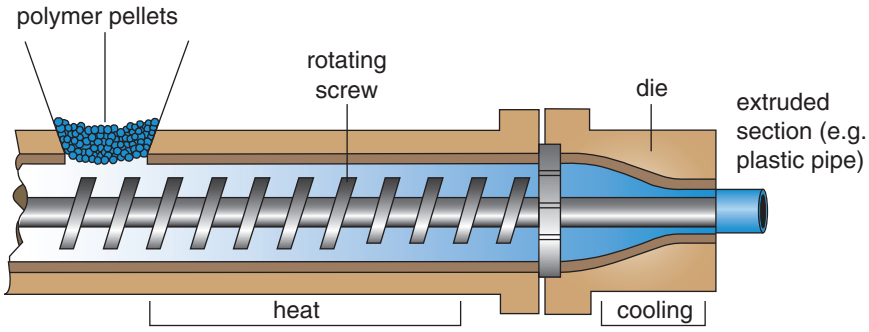


Fig. 50.31 Extrusion moulding process



Fig. 50.32 Products fabricated by extrusion moulding

The shape of the die determines the nature of the object that emerges from it. Pipe or tubing is extruded through a ring-shaped die formed around a central core.

Typical products fabricated by extrusion moulding include drain pipes, garden hoses, guttering, rods and unplasticized polyvinyl chloride (uPVC) window frames.

Reference Website
Refer to the following website for a video clip showing the process of vacuum forming:
https://www.youtube.com/watch?v=KGAuunWs8io
(accessed June 2015)

Vacuum forming

This method moulds objects from softened thermoplastic sheets. The sheet is positioned over a mould with small air canals. The air is then sucked out from underneath using a vacuum pump. Atmospheric pressure presses down on top of the plastic sheet and the sheet takes the shape of the mould.

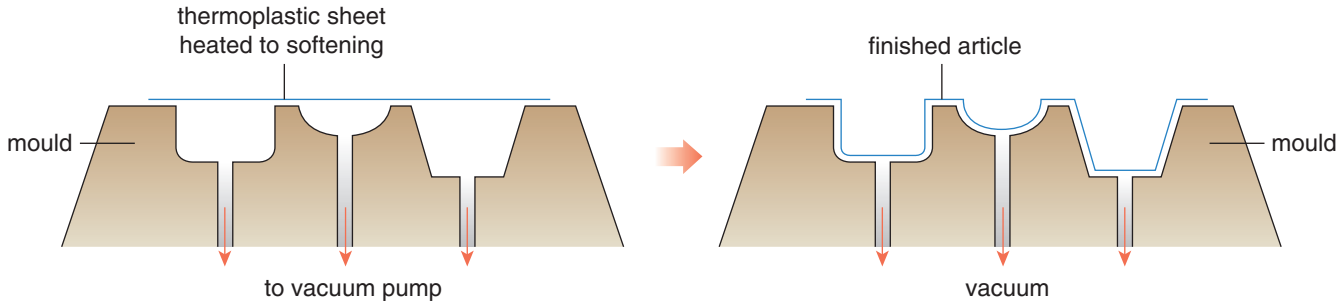


Fig. 50.33 Process of vacuum forming



Fig. 50.34 A chocolate box tray fabricated by vacuum forming

Typical products fabricated by vacuum forming include chocolate box trays, refrigerator linings, packaging trays, baths and acrylic sinks.

Compression moulding

Compression moulding is the most common method of forming thermosetting plastics. It is not generally used for thermoplastics.

Plastic moulding powder is put directly into the lower half of a heated mould. As the powder softens, the upper half of the mould is lowered to compress the melted plastic into shape. Continued heating induces a further polymerization. At this time, cross-links form between the polymer chains and the plastic sets.

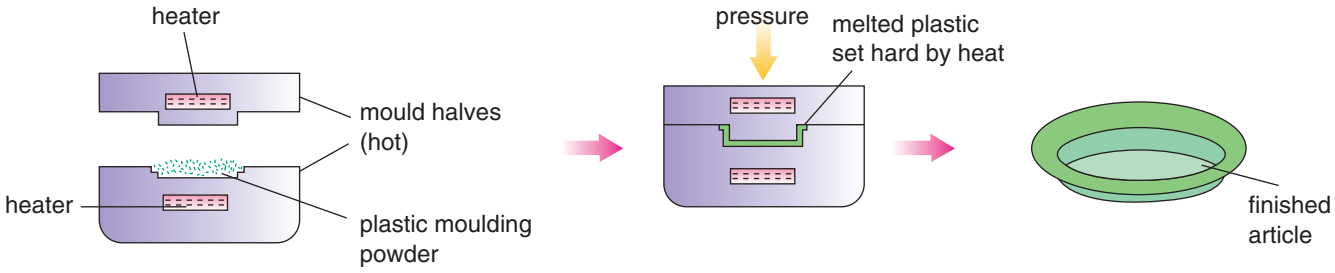


Fig. 50.35 Compression moulding process

Reference Website
Refer to the following website for an animation showing the process of compression moulding:
http://www.notesandsketches.co.uk/Compression_Moulding_DMC.html
(accessed May 2015)



Fig. 50.36 Products fabricated by compression moulding

Typical products fabricated by compression moulding include handles for cooking pots, children's tablewares, electric plugs, sockets and light switches.

Extension

50.8 Additives and plastics

Most plastics are blended with additives as they are processed into finished products. The additives are included to make plastics easier to process and to modify their working properties and product performance. N16

Below are types of common additive used:

- *antioxidants* — to prevent reaction with oxygen;
- *pigments* — to give colour;
- *stabilizers* — to inhibit undesirable degradation in use and prevent colours from fading;
- plasticizers — to make the plastics softer and easier to mould;
- blowing agents — to produce foamed materials;
- *fillers* — to make the products stronger and opaque.

Unplasticized polyvinyl chloride (uPVC) is rigid and hard. It is used to make window frames, gutters and piping.

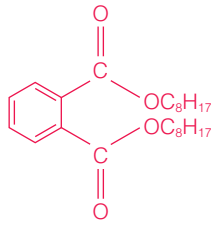
Plasticizers are one type of additive. When added to a polymer, the molecules of the plasticizer get in-between the polymer chains and allow the chains to slide past one another more easily. Products made of PVC with plasticizers are very flexible. PVC with plasticizers can be used to make articles such as shower curtains and wall coverings. N17 N18

ASL 2001 Q6(I)(c)(i)
ASL 2004 Q12(b)(II)(i)(2)
ASL 2006 Q13(b)(I)(iii)
AL 2010 Paper 1 Q10

N16 Examination questions may ask about the effect on the properties of a polymer upon the addition of various substances.

N17 Making PVC more flexible by the addition of a plasticizer – dioctyl phthalate (DOP)

Dioctyl phthalate (DOP) is an important plasticizer mainly used in the processing of PVC.



PVC contains polar C–Cl groups and DOP contains polar C=O groups. The polar groups of DOP are attracted towards the polar parts of the polymer chain. The non-polar groups of DOP prevents these polar parts from forming inter-polymer chain attractions. Thus, attractions between the polymer chains are reduced.

N18 Refer to page T39.

Extension

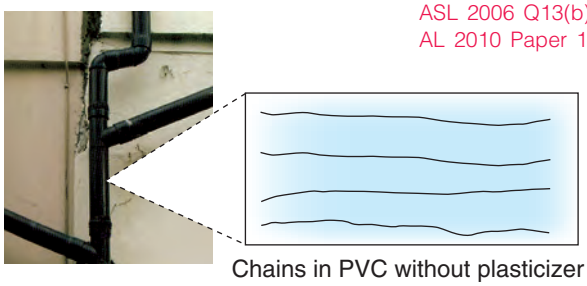


Fig. 50.37 Structures and uses of uPVC and PVC

antioxidant 抗氧化劑 pigment 染料 stabilizer 穩定劑 filler 填充劑

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

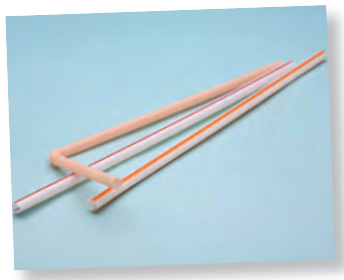
Suggest a suitable fabrication process for each of the following plastic products.



a) chair
Injection moulding



b) light switch
Compression moulding



c) plastic straw
Extrusion moulding



d) bottle
Blow moulding



e) acrylic sink
Vacuum forming



f) disposable drinking glass
Injection moulding

50.9 Recycling

N19 Refer to page T39.



Fig. 50.38 Plastic waste in a landfill

Plastics play an important part in our lives. However, excessive use of plastics creates environmental problems. In Topic 7 Fossil Fuels and Carbon Compounds, we have discussed four common methods for treating plastic waste, including N19

- landfilling;
- incineration;
- recycling; and
- pyrolysis.

We are going to look at recycling more closely here.

ASL 2003 Q6(a)
ASL 2010 Paper 1 Q12

Thermoplastics make up 80% of the plastics produced today. Recycling of thermoplastic waste* is important because

- it reduces the amount of plastic waste;
- it helps conserve petroleum which is non-renewable;
- it converts plastic waste into useful products.

* Thermosetting plastics decompose when heated, so they cannot be reheated and remoulded. This makes them difficult to recycle.

Single-plastic waste can be chopped into flakes, melted, and then remoulded. For example, HDPE milk jugs can be remoulded into detergent bottles, and PET beverage bottles can be turned into fibres, mainly for carpets. Single-plastic waste can also be broken down chemically into the monomer, and then re-polymerized.

Fig. 50.39 shows the life cycle of a plastic product that is recycled after use.

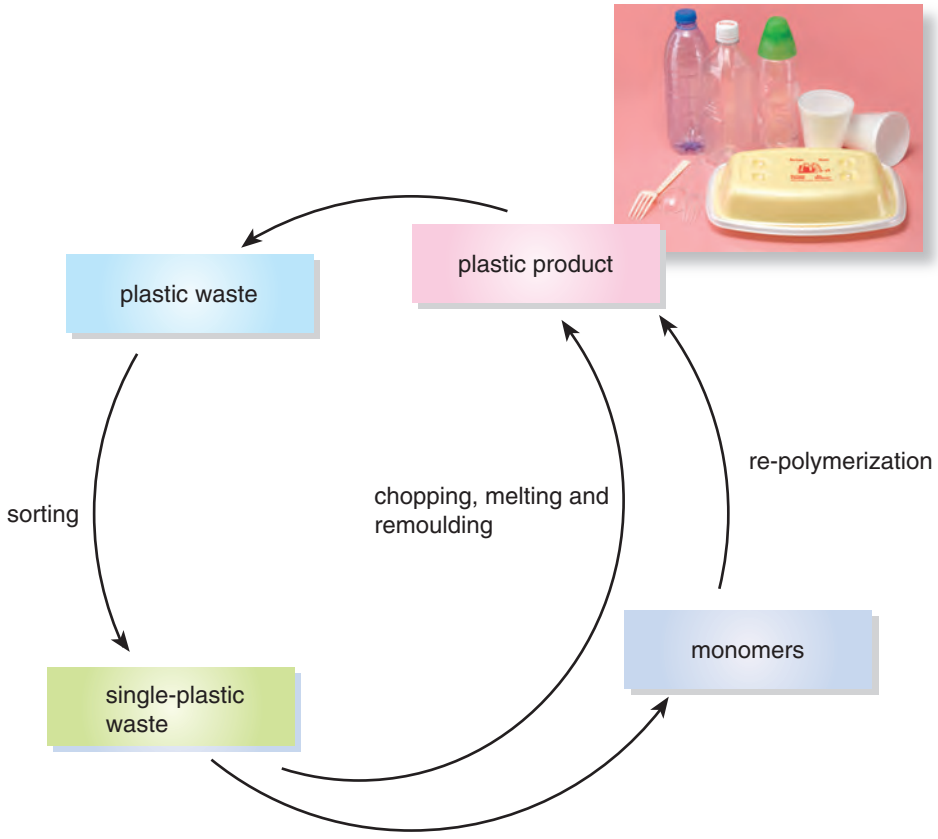


Fig. 50.39 Life cycle of a plastic product that is recycled after use

Reference Website
Refer to the following website for a video clip on plastic bottle recycling:
https://www.youtube.com/watch?v=zyF9MxlcItw
(accessed June 2015)



Fig. 50.40 This jacket is made of recycled plastic bottles

There is a wide range of products made from recycled plastics, such as low density polythene garbage can liners, PVC pipes and flooring, etc.

N20 Refer to page T40.

The Environmental Protection Department initiates the use of a plastic code system (Table 50.4). The system can help the sorting of different plastics in the recycling process. N20

Table 50.4 Major plastics and their uses

Code	Name of plastic	Examples of uses	Examples of recycled products
1 PET	poly(ethylene terephthalate)	beverage containers, ovenable pre-prepared food trays	detergent bottles, carpet fibres, fleece jackets
2 HDPE	high density polythene	milk jugs, detergent bottles, mixing bowls, toys, grocery bags	detergent bottles, recycling bins, drainage pipes, floor tiles
3 PVC	polyvinyl chloride	vegetable oil bottles, clear food packaging, pipes	tiles, plumbing pipe fittings, hoses
4 LDPE	low density polythene	films for packaging, sandwich bags, squeezable bottles, grocery bags	trash cans, grocery bags, landscape timber, garbage can liners
5 PP	polypropene	ketchup bottles, margarine tubs, medicine bottles, bottle crates	bins, flower pots, ice scrapers
6 PS	polystyrene	compact disc cases, cafeteria trays, egg cartons, disposable cups and plates	coat hangers, office accessories, CD boxes
7 OTHER	others	three and five gallons reusable water bottles, some citrus juice bottles	bottles, plastic lumber applications



Fig. 50.41 Plastic codes on plastic products

Worksheet
A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.



Discussion

Suggest problems associated with recycling.

- It is difficult to separate plastic waste from other waste.
- It is difficult to separate different plastic waste.
- Many plastics lose their original properties over successive recycling cycles.
- It is difficult to remove additives in the plastic waste.
- The process is uneconomical.

Worksheet
Two related worksheets with answers are available for download from *Jing Kung Chemistry Website*. Worksheet ① consists of additional questions for promoting active reading. (Note: Questions in this worksheet also cover those in the textbook.) Worksheet ②, requiring students to indicate the relationship among various terms by drawing, may help the less able readers improve their reading skills.



STSE Connections

Discovery of conductive polymers

We are used to the view that polymers do not conduct electricity. Yet researchers have changed this view with their discovery that a polymer, poly(ethyne), can be made conductive almost like a metal.

In the early 1970s, one of Shirakawa's students was trying to repeat the synthesis of poly(ethyne) in Japan. By mistake, he used 1000 times too much catalyst. He produced a silvery, metallic-looking film.

* Alan MacDiarmid, Hideki Shirakawa and Alan Heeger share the 2000 Nobel Prize in Chemistry for the discovery and development of conductive polymers.

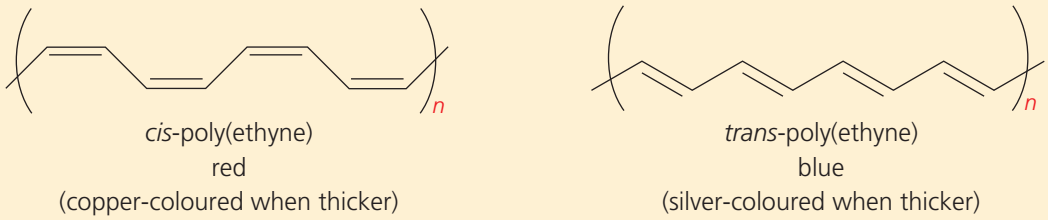
Alan MacDiarmid from the United States saw the film on a visit to the laboratory. He invited Shirakawa to work together to improve the electrical properties of the polymer. It was later found that with iodine treatment, the silvery material conducted electricity 1 billion times better than previous forms of poly(ethyne). Their studies opened up the field of 'plastic electronics'*.

The structure of poly(ethyne)

Ethyne is the simplest member of a family of unsaturated hydrocarbons called alkynes. The structure of ethyne is $H-C\equiv C-H$. When ethyne molecules polymerize, the product, poly(ethyne), contains molecules with alternating single and double bonds.

$$\cdots\cdots + H-C\equiv C-H + H-C\equiv C-H + H-C\equiv C-H + \cdots\cdots$$
$$\longrightarrow \text{-----} CH=CH-CH=CH-CH=CH \text{-----}$$

There is more than one form of poly(ethyne) as there are two ways in which the double bonds can be arranged relative to one another due to geometrical isomerism. The structures of two forms of poly(ethyne) are shown below. Of course, mixed *cis* and *trans* arrangements are also possible.



cis-poly(ethyne)
red
(copper-coloured when thicker)

trans-poly(ethyne)
blue
(silver-coloured when thicker)

Fig 50.42 Structures of *cis*-poly(ethyne) and *trans*-poly(ethyne)

In the *trans*-poly(ethyne), we can regard electrons forming the second bond in each double bond as being delocalized. Fig. 50.43 shows the structure of *trans*-poly(ethyne) with a dotted line representing the delocalized electrons. This isomer conducts electricity better than the *cis* isomer.



Fig. 50.43 Schematic representation of the delocalized electrons of *trans*-poly(ethyne)

Fig. 50.44 shows how the conductivity of *trans*- and *cis*-poly(ethyne) compared with that of other materials.

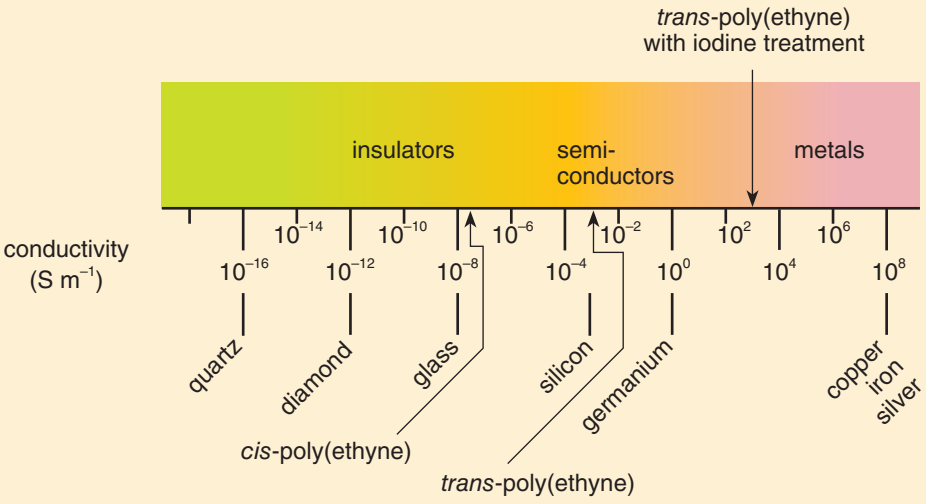


Fig 50.44 Conductivity of conductive polymers compared to those of other materials, from quartz (insulator) to copper (conductor)

Applications of conductive polymers

By coating an insulator with a very thin layer of conducting polymer, it is possible to prevent the buildup of static electricity. This is particularly important where a discharge of static electricity is undesirable. Such a discharge can be dangerous in an environment with flammable gases and liquids and also in the explosives industry. In the computer industry, the sudden discharge of static electricity can damage the microcircuits.

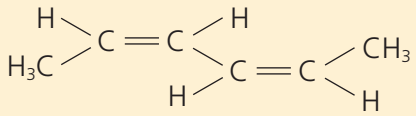
By placing the monomer between two conducting surfaces and allowing the monomer to polymerize, it is possible to stick the conducting surfaces together. The polymer is a conductive adhesive and allows an electric current to pass through it.

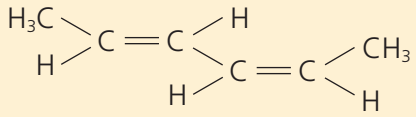
Many electrical devices, particularly computers, generate electromagnetic radiation. This can cause malfunctions in nearby electric devices. This radiation can be absorbed by coating the inside of the plastic casing of a computer with a conductive surface. This can best be achieved by using conducting polymers. They are cheap, easy to apply and can be used with a wide range of plastics.

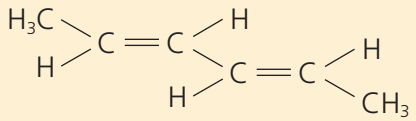
Questions Refer to page T41.

1 Suggest TWO reactions in which ethene and ethyne behave similarly.

2 Three possible structures of hexa-2,4-diene are shown below. Identify the one with a mixed *cis* and *trans* arrangement.







Summary

1 The following table compares the structures and properties of low density polythene and high density polythene.

Polythene	Structure	Properties
Low density polythene (LDPE)	• contains relatively short chains with a considerable degree of branching • chains cannot lie close together • van der Waals' forces between molecules are smaller	• lower density • lower tensile strength • lower melting point
High density polythene (HDPE)	• contains longer chains that are not branched • chains can lie close together • van der Waals' forces between molecules are greater	• higher density • higher tensile strength • higher melting point

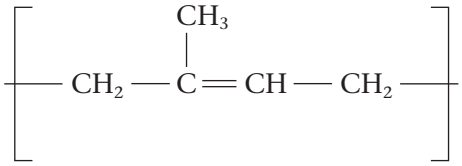
2 a) Hydrogen bonds exist between the polymer chains of Kevlar.

b) As there are benzene rings in the polymer chains of Kevlar, the chains are very rigid and linear.

c) The hydrogen bonds cause the polymer chains to interlock with each other.

d) The strength of Kevlar is greater than that of nylon.

3 a) The repeating unit of natural rubber is



b) Vulcanization improves the strength and elasticity of rubber by cross-linking individual polymer molecules.

4 a) Degradation of a plastic is defined as a harmful change in its appearance, physical properties or chemical structure.

b) Biodegradable plastic is a degradable plastic in which the degradation results from the action of naturally occurring microorganisms, such as bacteria, fungi, etc.

c) Two major directions of developing degradable plastics are:

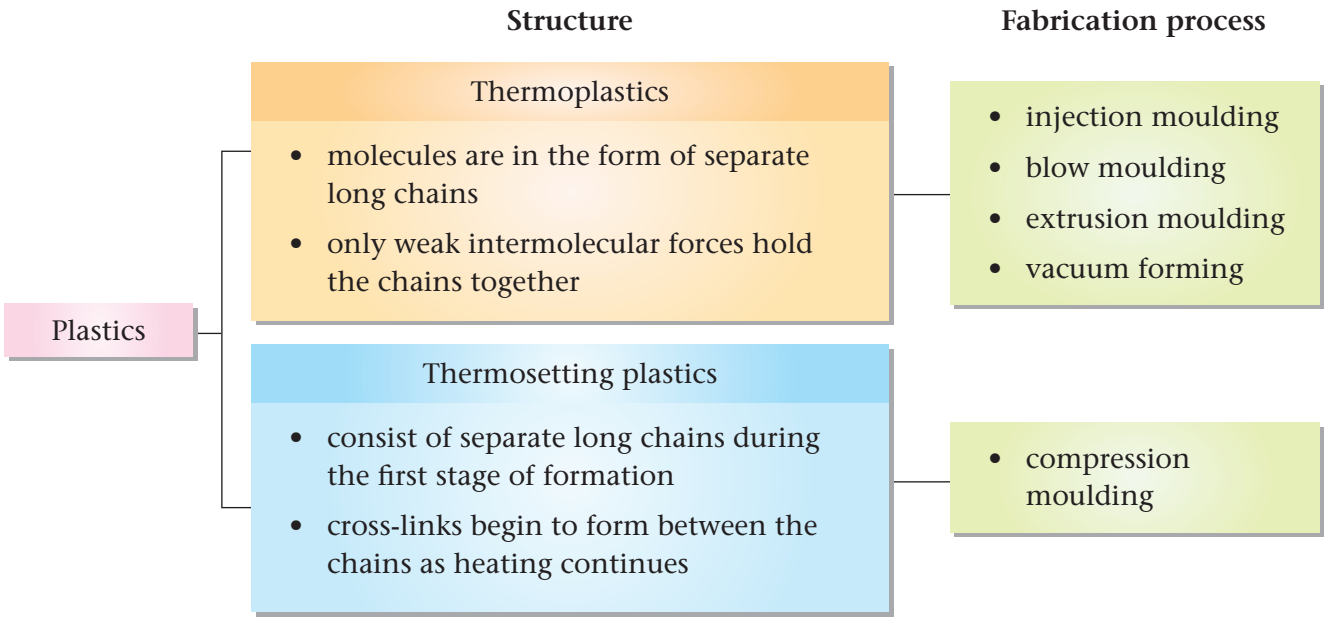
i) modifying petroleum-based plastics with additives;

ii) making plastics from plant-based materials.

5 a) Polylactide is a biodegradable polyester whose monomer is lactic acid.

b) Lactic acid can be obtained from renewable resources, such as corn.

6 The following diagram summarizes information about the structures and fabrication processes of thermoplastics and thermosetting plastics.



7 Four common methods for treating plastic waste are:

a) landfilling;

b) incineration;

c) recycling; and

d) pyrolysis.

Exercise

Note: The symbol 🌶 indicates the level of difficulty of a question.

Part I Knowledge and understanding

1 Construct a concept map for the concepts learnt in this unit.
Refer to page T41.

2 A student investigates the properties of five types of polymer, A, B, C, D and E.

polymer	flexibility	melting point
A	flexible	213 °C
B	flexible	110 °C
C	hard and brittle	starts to burn before it melts
D	flexible	334 °C
E	hard and brittle	starts to burn before it melts

a) Which of the polymers has the weakest forces between its molecules?
B

b) Which TWO polymers have cross-links between their molecules?
C and E

3 Most plastics are blended with additives as they are processed into the finished products.

Extension

a) Suggest THREE additives commonly added to improve the properties of plastics.

b) Give the function of each additives suggested in (a).

c) Describe the fabrication process involved in the production of packaging trays.
Vacuum forming

4 Name FIVE common plastic fabrication processes.
Refer to page T41.

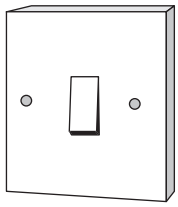
5 Give THREE problems associated with the recycling of plastics.
Refer to page T42.

3 (a) & (b) Any three of the following:
- Antioxidants — to prevent reaction with oxygen.
- Pigments — to give colour.
- Stabilizers — to inhibit undesirable degradation in use and prevent colours from fading.
- Plasticizers — to make the plastics softer and easier to mould.
- Blowing agents — to produce foamed materials.
- Fillers — to make the products stronger and opaque.

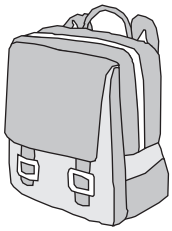
Part II Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

6 The following diagrams show some items made of plastics.



electric switch



rucksack



shirt

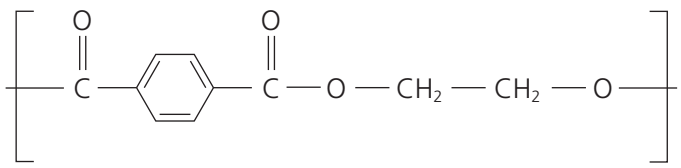
c) Compression moulding
Plastic moulding powder is put directly into the lower half of a heated mould.
As the powder softens, the upper half of the mould is lowered to compress the melted plastic into shape.
Continued heating induces a further polymerization. At this time, cross-links form between the polymer chains and the plastic sets.

a) Which of the above items is most likely to be made of a thermosetting plastic? Explain your answer. Electric switch
It should be able to withstand high temperatures.

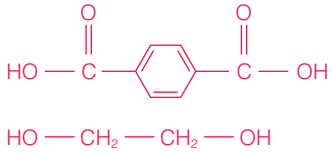
b) Name one plastic which is suitable for making the rucksack.
Nylon

c) Describe the fabrication process involved in the production of the electric switch.

d) A polyester is used to make the shirt. The repeating unit of the polyester is shown below:



d) i)



i) Draw the structural formulae of the monomers of the polyester.

ii) Name the type of polymerization involved in the production of the polyester.
Condensation polymerization

iii) What would be observed when a drop of concentrated solution of an alkali is dropped onto the shirt? Explain your answer.
A hole appears because the alkali hydrolyzes the ester linkage in the polyester.

7 Polythene is obtained from ethene. The table lists the properties of two types of polythene obtained when using different reaction conditions.
Refer to page T42.

Process	Reaction conditions	Information of polythene obtained
I	temperature: 100 – 300 °C pressure: 1000 – 2000 atm catalyst	• more branch polymer chains in the polythene • density of polythene: 0.92 – 0.94 g cm^{-1}
II	temperature: 30 – 80 °C pressure: 5 – 30 atm catalyst	• fewer branch polymer chains in the polythene • density of polythene: 0.94 – 0.96 g cm^{-1}

a) Write a chemical equation for the polymerization involved.

b) Name the type of polymerization involved.

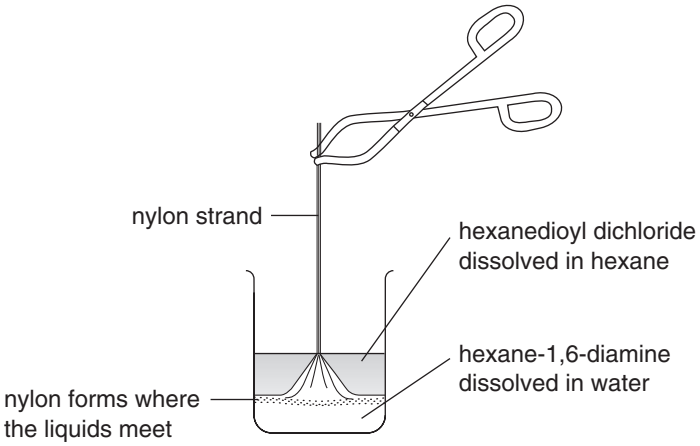
c) Based on the above information, explain which type of polythene is more costly to produce.

d) Suggest which type of polythene has polymer chains that are more tightly packed. Explain your answer.

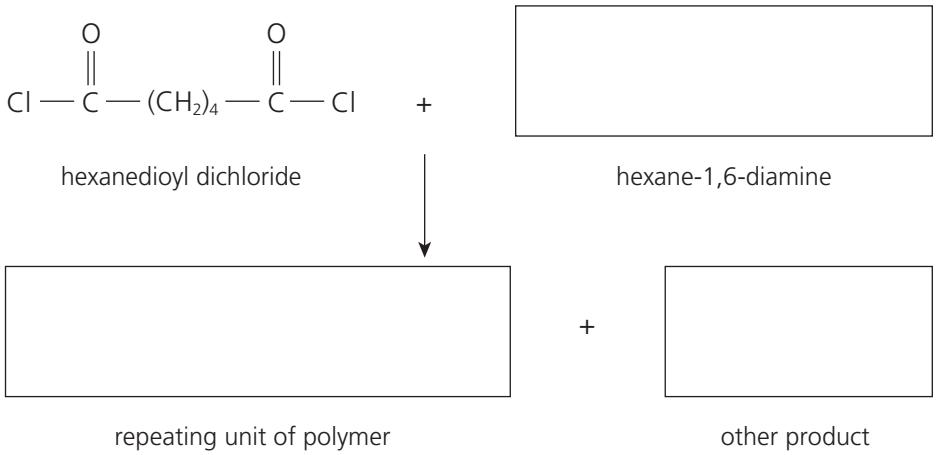
e) Briefly describe the fabrication process involved in the production of food boxes from polythene beads.

8 The 'Nylon Rope Trick' is a well known laboratory demonstration for the formation of a condensation polymer from its monomers.

Two solutions, one containing each monomer, are placed in a beaker as shown. In this reaction, the more reactive hexanedioyl dichloride is used instead of hexane-1,6-dioic acid. The nylon forms where the two immiscible liquids join, and can be pulled out from between the layers in a continuous strand.
Refer to page T42.

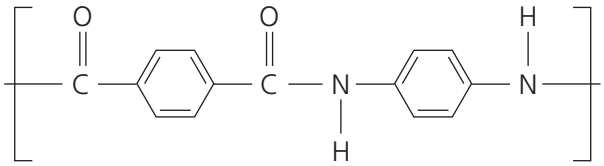


a) Complete the equation below to show the formation of the nylon polymer from its monomers. Show a repeating unit for the polymer and the other product formed.



b) Nylon is sometimes used for electrical insulation. However, if there is a risk of high temperatures then a polymer such as Nomex, with a higher melting point, is used.

The repeating unit of Nomex is shown below.

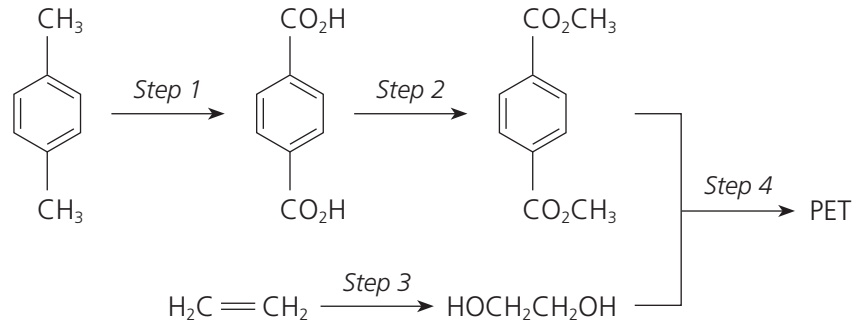


i) Draw the structures of two monomers that could be used to form Nomex.

ii) Suggest a reason why the melting point of Nomex is higher than that of nylon.

(OCR Advanced GCE, Chains, Rings and Spectroscopy, Jun. 2007)

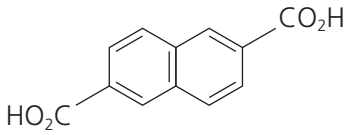
9 Poly(ethylene terephthalate) (PET) is a polymer used in making clothing fibres and soft drink bottles. The flow diagram below shows a route for the synthesis of PET.



a) Name the type of reaction for the transformation in *Step 1*.

b) Give the reagent(s) and conditions for each of *Steps 2*, *3* and *4*.

c) Poly(ethylene naphthalate) (PEN) is a polymer formed from naphthalic acid and ethane-1,2-diol.



naphthalic acid

i) Draw the structure of a repeating unit of PEN.

ii) Which polymer, PET or PEN, do you expect to be more rigid? Explain.

(HKASLE 2007)

10 Kevlar is a polymer made from the following two monomers:
Refer to page T43.



a) Draw the repeating unit of Kevlar.

b) Name the type of polymerization involved in the production of Kevlar.

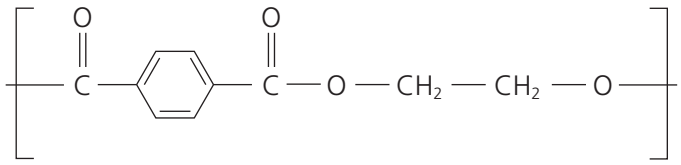
c) Draw a diagram to show how polymer chains in Kevlar are attracted to each other. Hence explain why Kevlar has a high strength.

d) On many offshore oil drilling platforms, steel ropes have been replaced by Kevlar ropes. Apart from strength, state TWO advantages of using Kevlar ropes on oil drilling platforms.

e) Apart from the use in ropes, suggest ONE other application of Kevlar.

11 X-ray film is made by coating a plastic sheet with a silver halide. The plastic sheet is made from a mixture of a polyester and a nylon.
Refer to page T44.

a) A typical polyester used in the making of films is PET. A repeating unit is shown below.



PET

Draw the full structural formula of the alcohol used as a monomer to make PET.

b) A nylon can be made by polymerizing a mixture of the monomers D and E.

$HOOC(CH_2)_4COOH$ $H_2N(CH_2)_6NH_2$

D E

i) Give the systematic name of compound E.

ii) Nylon polymers are formed by a condensation reaction. Use the reaction between compounds D and E to explain the meaning of condensation.

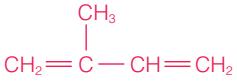
c) Nylon is added to polyester to make X-ray film stronger. Explain, in terms of intermolecular attractions, why the addition of nylon strengthens the film.

d) i) Large amounts of used film are disposed of every year. After the silver halide has been removed from the film, the remaining polymer sheets can be disposed of by burning or by burying them in the ground. Give ONE advantage for each method relative to the other.

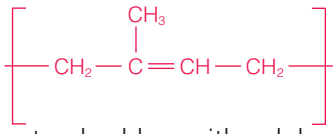
ii) Chemists are developing methods to convert some waste polymers into useful chemicals.

Polyesters can be broken down by hydrolysis into organic acids and alcohols. Give the conditions and reagents used for hydrolyzing an ester link.

(OCR Advanced GCE (Salters), Chemistry of Materials, Jun. 2005)

12 Natural rubber is a polymer of 2-methylbuta-1,3-diene. 

a) Draw the structural formula of 2-methylbuta-1,3-diene.

b) Draw the repeating unit of natural rubber. 

c) Natural rubber has poor strength and elasticity. Heating natural rubber with sulphur greatly improves these properties of rubber.

i) Name the process used.
Vulcanization

ii) What feature of the polymer chain of rubber enables this process to take place?
The double bonds in the polymer chains allow addition reactions to occur.

iii) Explain why the process improves the strength of rubber.
The sulphur reacts with some of the double bonds in each chain, forming cross-links of one to about four sulphur atoms between the chains. This three-dimensional cross-linked structure makes vulcanized rubber stronger.

13 Plastics can be classified into thermoplastics and thermosetting plastics according to their thermal properties.

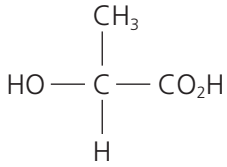
a) Explain, in terms of bonding and structure, why thermoplastics and thermosetting plastics behave differently upon heating.

b) Polythene (PE) is a thermoplastic commonly used in making shopping bags. Such bags are durable.

i) Write the chemical equation for the formation of PE from its monomer.

ii) Explain, in terms of bonding, why objects made of PE are durable.

c) Some scientists suggested using polylactic acid (PLA) instead of PE to make shopping bags as this can help reduce environmental problems. PLA is a polyester made from lactic acid, which has the following structure:



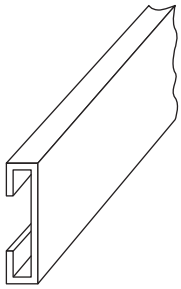
i) Identify the functional groups in lactic acid.

ii) Write the repeating unit of PLA.

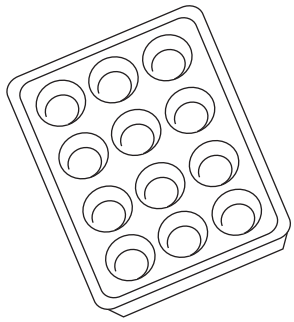
iii) Explain, in chemical terms, why the disposal of PLA poses less harm to the environment than that of PE.

(HKCEE 2006)

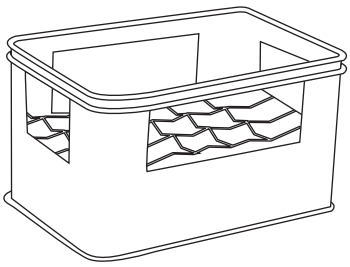
14 Suggest a suitable fabrication process for each of the following plastic products.



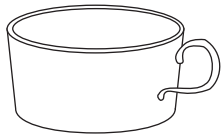
a) curtain rail
Extrusion moulding



b) chocolate box tray
Vacuum forming



c) bottle crate
Injection moulding

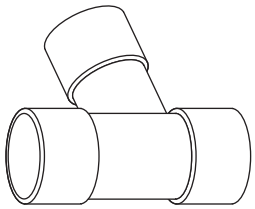


d) a thermosetting plastic soup cup
Compression moulding

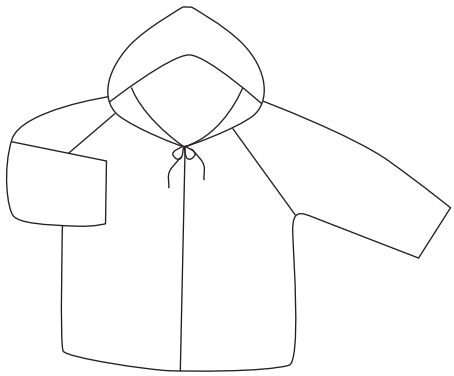


e) feeding bottle
Blow moulding

15 Two articles made of polyvinyl chloride (PVC) are shown below.



water pipe



raincoat

a) Polyvinyl chloride is suitable for making the water pipe because it is hard and rigid. Explain why it is hard and rigid in terms of intermolecular forces. There are polar-polar interactions between the PVC polymer chains. The chains cannot slide past each other easily. Hence PVC is hard and rigid.

b) Briefly describe the fabrication process involved in the production of the water pipe.
Plastic pellets are loaded into a hopper, then fed into a long heating chamber.
At the end of the heating chamber, the molten plastic is forced out through a ring-shaped die formed around a central core.

15 c) Add plasticizer to the polymer.
The molecules of the plasticizer get in-between the polymer chains and allow the chains to slide past one another more easily.
Thus, the raincoat made is less rigid.

c) Why can the raincoat be made less rigid than the water pipe?

d) i) Any one of the following:
- Causes acid rain.
- Attacks the respiratory system.

Extension

d) Incineration of PVC wastes produces hydrogen chloride which causes air pollution.

i) State ONE harmful effect of the discharge of hydrogen chloride into the atmosphere.

ii) Suggest how hydrogen chloride can be removed from incinerator's flue gas prior to its discharge to the atmosphere.
Install scrubbers

16 The polymer commonly known as PVC exists in two forms. Plasticized PVC is used where flexibility is required. Unplasticized PVC, uPVC, is rigid at room temperature and is used to make things such as guttering for houses.

a) Suggest ONE other use of uPVC, connected with a house.

Any one of the following:
- Drain pipes
- Window frames

b) PVC is manufactured by polymerizing 'vinyl chloride'. Vinyl chloride is produced in a two stage synthesis as outlined below.

Stage 1 $CH_2=CH_2 + Cl_2 \xrightarrow{FeCl_3} ClCH_2CH_2Cl$

Stage 2 $ClCH_2CH_2Cl \xrightarrow{heat} \underset{\text{vinyl chloride}}{CH_2=CHCl} + HCl$

i) Give the systematic name for vinyl chloride.
Chloroethene

ii) Underline ONE of the following words to describe the reaction in *Stage 1*.

addition elimination substitution

iii) Select ONE word from the list to describe the reaction in *Stage 2*.
Elimination

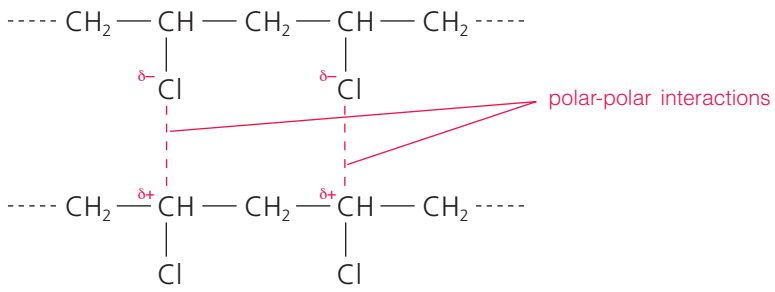
iv) What type of polymerization is the polymerization of vinyl chloride?
Addition polymerization

Extension

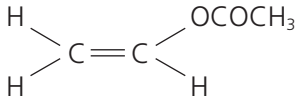
c) PVC owes many of its properties to the intermolecular forces between the polymer chains.

i) Name the strongest type of intermolecular force that is present in PVC.
Polar-polar interactions

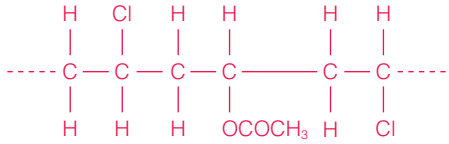
ii) Use the diagram below to show how these intermolecular forces hold the PVC chains together.



d) Bumpers and spoilers on cars are made from plasticized PVC. One way of making PVC more plastic is to incorporate another monomer unit into the polymerization process. A monomer used in this way is ethenyl ethanoate.



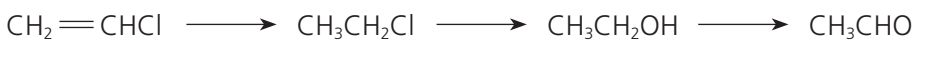
ethenyl ethanoate



Draw out a section of the polymer formed from vinyl chloride and ethenyl ethanoate. Include TWO units of vinyl chloride and ONE unit of ethenyl ethanoate.

e) When ethenyl ethanoate is incorporated, the polymer becomes more flexible. This is because the polymer chains do not pack together so well. Explain why this leads to greater flexibility.
The polar-polar interactions between polymer chains become weaker when working over longer distance.

f) Vinyl chloride will also undergo the following sequence of reactions.



vinyl chloride — chloroethane — ethanol — compound A

i) Name the reagent and conditions needed to turn vinyl chloride into chloroethane. Hydrogen
Platinum / palladium catalyst

ii) Classify ethanol as primary, secondary or tertiary, giving a reason. Primary
The carbon atom bearing the –OH group has only one alkyl group attached to it.

iii) Name the functional group in compound A.
Carbonyl group

iv) Give the reagents and conditions for the conversion of ethanol to compound A in the laboratory. Heat ethanol with potassium dichromate solution and dilute sulphuric acid.
Distil off compound A once formed.

(OCR Advanced Subsidiary GCE (Salters), Chemistry of Natural Resources, Jan. 2006)

🌶🌶🌶 17 Drainage pipes and shower curtains are usually made of plastics nowadays.

a) Name a plastic which can be used to make both of these items. Draw the repeating unit of the polymer in this plastic.

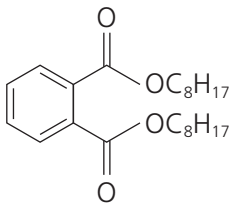
b) Suggest a moulding method to make

i) drainage pipes.

ii) shower curtains.

Exercise

c) Dioctyl phthalate (DOP) can be added to the plastic in (a) rendering it suitable for making shower curtains. DOP has the following structure:

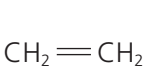


State the function of DOP. Explain your answer in terms of intermolecular forces.

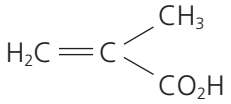
d) Is it appropriate to dispose of abandoned drainage pipes by incineration? Explain your answer.

(HKASLE 2006)

18 Ethene and 2-methylpropenoic acid react in the presence of benzoyl peroxide to give polymer N, which does not have a definite repeating unit.

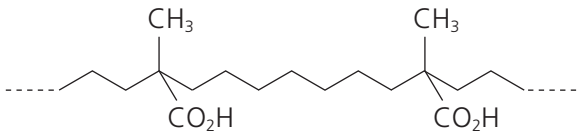


ethene



2-methylpropenoic acid

A portion of the structure of N is shown below:



a) Name the type of polymerization involved in the formation of N.

b) State, with explanation, whether or not N is a thermosetting polymer.

c) Explain why the strength of N obtained depends on the relative amounts of ethene and 2-methylpropenoic acid used in the polymerization.

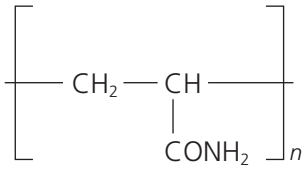
d) N reacts with excess sodium hydroxide to give another polymer P.

i) Draw the structure of P which corresponds to the portion of the structure shown above.

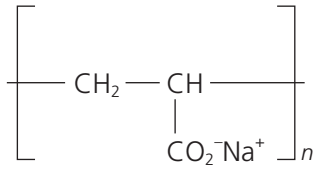
ii) Compare the strengths of N and P. Explain your answer.

(HKALE 2004)

19 A brand of baby diaper uses polyacrylamide and sodium polyacrylate as water absorbing materials. The structures of the two polymers are shown below:



polyacrylamide



sodium polyacrylate

a) Draw the structure of the monomer of polyacrylamide.

b) Suggest a synthetic route, with NOT MORE THAN THREE STEPS, for the transformation of propenoic acid to sodium polyacrylate.

c) Account for the water absorbing property of the following materials:

i) polyacrylamide

ii) sodium polyacrylate

d) Apart from their use in diapers, suggest ONE other application of such water absorbing materials in daily life.

(HKASLE 2005)

Exercise

Extension

20 Biodegradable polymers are environmentally friendly. Write an article for your school magazine to promote more extensive use of biodegradable polymers in Hong Kong.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

(HKALE 2004)

21 Discuss how bonding and structure of polymers affect their physical properties.

(For this question, you are required to give answers in paragraph form. Use equations, diagrams and examples where appropriate.)

(HKASLE 2007)